

COMPARATIVE ANALYSIS OF PIEZOELECTRIC MATERIALS IN A SANDWICH AIRCRAFT WING

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COMPARATIVE ANALYSIS OF PIEZOELECTRIC MATERIALS IN A SANDWICH AIRCRAFT WING

*A Project Report
submitted in partial fulfilment of the
requirements for the award of the degree of*

**Bachelor of Technology
in
Aeronautical Engineering**

by

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April, 2027

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ACKNOWLEDGEMENT

The satisfaction that accompanies the successful completion of any task would be incomplete without the mention of the people who made it possible and whose constant guidance and encouragement crown all the efforts with success. we thank our college management and **Sri. M. Rajashekhar Reddy**, Chairman, for providing us the necessary infrastructure to carry the project work.

We express sincere thanks to **Dr. L V Narasimha Prasad**, the beloved Principal who has been a great source of information for our work, and to **Dr. CH Sandeep**, Professor and Head, Department of Aeronautical Engineering, for extending his support to carry out the project work.

We are especially thankful to our internal supervisor, **Dr. BDY Sunil**, Professor, Aeronautical Engineering, for the successful project.

We take this opportunity to express our thanks to one and all who directly or indirectly helped us in bringing this effort to present form.

ABSTRACT

With the growing demand for lighter, multifunctional and smart structures, the concept of vibration energy harvesting has gained much importance. The wings of the aircraft are always under stress due to aerodynamic forces and vibrations. The energy generated from such vibrations could be used as a source of electrical energy for various applications including Structural Health Monitoring (SHM). This project focuses on the use of piezoelectric materials in a sandwich aircraft wing to study its structural, dynamic and energy harvesting characteristics.

The model for the analysis of a three-dimensional sandwich aircraft wing having a piezoelectric material layer was created using CATIA V5 and analyzed using ANSYS Mechanical 2025 R2 software. Five commercially available PZT materials such as PZT-4, PZT-5A, PZT-5H, PZT-8, and PZT-26 have been comparatively studied. All the wings under study have been kept similar in terms of geometry, material, meshing, contacts, loading, and boundary conditions. Static structural analysis is done to find out the total displacement, equivalent von Mises stress, and elastic strain. Modal analysis is done to find out the natural frequencies and modes.

According to the findings, PZT-26 had the least static deflection and the highest natural frequency of 54.193 Hz. The highest electric potential was achieved by PZT-8 at 6275.1 mV. In general, PZT-26 was found to be superior in terms of its structural properties, dynamism, and energy generation abilities, making it the optimal material for the suggested smart aircraft wing design.

Keywords: Piezoelectric Materials, PZT, Vibration Energy Harvesting, Finite Element Analysis, Modal Analysis, Coupled-Field Harmonic Analysis, Structural Health Monitor.

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LIST OF ABBREVIATIONS

ABS	Acrylonitrile Butadiene Styrene
ANSYS	Analysis System
BaTiO ₃	Barium Titanate
CAD	Computer-Aided Design
CATIA	Computer-Aided Three-Dimensional Interactive Application
CNT / CNTs	Carbon Nanotubes
FEA	Finite Element Analysis
FEM	Finite Element Method
FSI	Fluid-Structure Interaction
Hz	Hertz
IARE	Institute of Aeronautical Engineering
kg	Kilogram
m	Meter
MEMS	Micro-Electro-Mechanical Systems
MFC	Macro Fiber Composite
mm	Millimeter
MPa	Megapascal
mV	Millivolt
Pa	Pascal
PLA	Polylactic Acid
PVDF	Polyvinylidene Fluoride
PZT	Lead Zirconate Titanate
SHM	Structural Health Monitoring
ZnO	Zinc Oxide

Chapter 1 Introduction

1.1 Introduction

The development of aerospace technology has resulted in a rising need for lightweight, efficient, and intelligent structures that enhance the performance of the aircraft and reduce the amount of maintenance required. Not only must modern airplanes survive complex aerodynamic and mechanical loads but they should also be able to perform additional tasks, including structural monitoring and energy management. As a result, smart structures which incorporate sensing, actuation, and energy harvesting functionality are highly researched.

One of the energy harvesting methods, vibration energy harvesting based on piezoelectric materials, is especially promising since it directly transforms mechanical strain into electrical energy. Due to the constant vibration of the airplane structure in operation, the otherwise wasted mechanical energy can be used to supply energy for low-power electronic devices and structural health monitoring systems[1]. The integration of piezoelectric elements in light sandwich panels allows for load-bearing and electricity generation without much additional mass in the structure. These smart multifunctional structures will help create self-powered structures with better reliability and less need for maintenance[2].

The current study focuses on a piezoelectric sandwich plate of the aircraft wing skin for energy harvesting based on vibration. Numerical analysis is done to analyze the structural response and electricity generation capability of the piezoelectric structure.

1.2 Background

The increased requirement for sustainable engineering systems has boosted research into energy harvesters, which use ambient energy and transform it into electrical energy. Among other types of energy, mechanical vibration is especially promising for aerospace applications because aerospace structures experience constant exposure to aerodynamic loading, engine vibrations, and turbulent flows[1].

As opposed to some other harvesting techniques, piezoelectric harvesting provides a number of benefits such as efficient energy conversion, small size, easy integration, and minimum maintenance. When used in sandwich composite structures, piezoelectric materials allow developing self-powered systems that can generate electricity for monitoring sensors and other devices[1]. Energy-producing aerospace structures reduce reliance on traditional batteries as well as increasing the reliability of the system and decreasing maintenance needs. Parts of the

airplane, such as the wings, fuselage plates, control surfaces, and engine mounting, are under continuous vibration; therefore, they provide a perfect opportunity to install vibration energy harvesters. Incorporating piezoelectric materials in aerospace structures helps build light, intelligent, and energy-efficient airplanes[2], [3].

1.3 Piezoelectric Materials

The piezoelectric materials are classified as smart materials which can convert mechanical energy to electrical energy and vice versa using electromechanical coupling. It is due to this reason that piezoelectric materials find many applications in areas such as sensing, actuation, vibration control, health monitoring, and vibration energy harvesting. The efficient energy conversion capability, small size, and ease of use with engineering structures make them highly useful for aerospace applications[1].

There are many types of piezoelectric materials that are used in engineering applications, depending upon the required electromechanical behaviour, strength, flexibility, and working conditions of the application. Table 1.1 provides details about common piezoelectric materials along with their main characteristics and applications.

Table 1.1 : Comparison of commonly used piezoelectric materials and their applications

Material	Main Characteristics	Typical Applications	Suitability for Energy Harvesting
Lead Zirconate Titanate (PZT)	High piezoelectric coefficient, excellent electromechanical coupling and high energy conversion efficiency	Energy harvesting, sensors, actuators and structural health monitoring	Excellent
Polyvinylidene Fluoride (PVDF)	Flexible, lightweight and corrosion resistant	Flexible sensors and wearable electronics	Moderate
Barium Titanate (BaTiO ₃)	Lead-free ceramic with good dielectric properties	Sensors, capacitors and electronic devices	Moderate
Zinc Oxide (ZnO)	Suitable for micro- and nano-scale devices	MEMS and nanosensors	Moderate

Macro Fiber Composite (MFC)	Flexible composite material with good conformability	Smart structures, vibration control and aerospace applications	Good
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Piezoelectric ceramics, in general, show more electromechanical coupling as well as energy conversion properties as compared to flexible polymers. Even though PVDF and MFC are suitable for use in cases where structural flexibility is desired, PZT ceramics can produce better electrical response and have excellent energy harvesting abilities. Therefore, PZT is one of the most extensively used materials for vibration energy harvesting from aerospace applications and structural health monitoring due to its combination of reliability and high efficiency[1], [3]. Due to the existence of many commercial grades of PZT with varying electrical and mechanical properties, in this research, more than one grade of PZT material has been considered. Table 1.2 illustrates the five types of piezoelectric ceramics that have been chosen for the numerical analysis along with the reasons for selecting them.

Table 1.2 : Piezoelectric materials selected for the present numerical investigation

Material	General Characteristics	Reason for Selection
PZT-4	High stiffness and good mechanical strength	Selected to evaluate structural stability under vibration loading.
PZT-5A	Balanced mechanical and piezoelectric properties	Represents a widely used engineering-grade piezoelectric ceramic suitable for aerospace structures.
PZT-5H	High piezoelectric coefficient and high sensitivity	Selected to investigate the influence of high electromechanical coupling on electrical energy generation.
PZT-8	High mechanical quality factor and low dielectric loss	Suitable for continuous operation and high-power applications.
PZT-26	Good mechanical stability and reliable electromechanical performance	Included to compare its structural response and energy harvesting capability with other commercial PZT materials.

The chosen materials provide a wide variety of commercially available PZT ceramics with different electromechanical properties, allowing for fair comparison under similar loading and

boundary conditions. Their performance will be tested using the same aircraft wing sandwich plate structure, so that any difference in terms of structural performance and electricity generation can be related only to the material differences. Material properties applied to the FEM model will be provided in [Chapter 6](#), while the comparative performance in terms of structure and energy generation will be described in [Chapter 7](#)[2], [3].

1.4 Sandwich Structures

Sandwich structures are light structural designs with two stiff face sheets joined together using a core that is light in weight. The combination results in high bending stiffness and strength despite the lightweight design of sandwich panels that are utilized in aerospace, automotive, maritime, and civil engineering industries. The stiffness to weight ratio and load transfer efficiency have made them ideal for application on aircraft structures[4].

The structural design of a sandwich panel is adjustable depending on the type of materials to use as face sheets, core geometry, and the material for the core to meet special requirements. In the current era, the introduction of smart materials like piezoelectric material into sandwich structures has widened the use of sandwich panels through vibration control, health monitoring, and vibration-based energy harvesting. As an example of the basic construction used in this research, the design of a sandwich panel is shown in Figure 1.1 below.

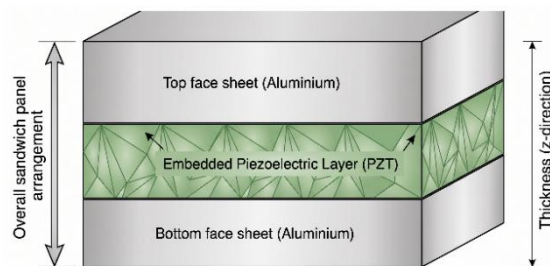


Figure 1.1 : Basic sandwich structure with an embedded piezoelectric layer.

This layered structure makes the face sheets capable of handling tensile and compression forces, whereas the core handles shear forces and keeps a gap between the two external layers. This greatly increases the stiffness of the component in a much more efficient way without adding too much weight[4]. Due to the many benefits offered by sandwich structures, they are used very frequently in contemporary airplanes. Lightweight construction saves fuel, whereas the core provides stiffness and impact resistance. The main features that make them popular are summarized in Table 1.3 below.

Table 1.3 : Major advantages of sandwich structures

Advantage	Engineering Benefit
High strength-to-weight ratio	Reduces structural weight while maintaining strength
High bending stiffness	Improves load-carrying capability
Good vibration damping	Enhances structural stability
Excellent energy absorption	Improves impact resistance
Design flexibility	Allows different face sheet and core combinations
Multifunctional capability	Supports integration of sensors and piezoelectric materials

Due to their low weight, high rigidity, and multifunctionality, sandwich panels have been widely used as one of the best choices for aircraft structural elements such as aircraft wings, fuselage panels, control surfaces, radomes, and interior parts. The capacity of these panels to incorporate sensors and piezoelectric materials makes it possible to design smart structures capable of structural monitoring and vibration energy harvesting[2]. The aircraft wing studied in the current research is modeled using the sandwich panel consisting of embedded piezoelectric material to examine vibration energy harvesting[2], [5].

1.5 Motivation of the Present Work

Wings of aircrafts are always exposed to the loads created due to aerodynamic flow, engine excitation, turbulence, and vibrations during the entire process of flight. While vibrations have been regarded as undesirable, since they affect not only structural behavior but also passenger comfort, they are actually one of the sources of mechanical energy that can be harnessed through vibration-based energy harvesting[2]. Vibration modes that aircraft wings usually experience are represented in Figure 1.2 below.

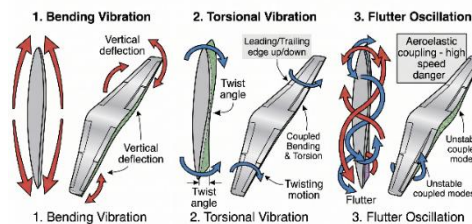


Figure 1.2 : Typical vibration modes experienced by an aircraft wing.

With the rising utilization of SHM systems, there has been a further increase in the need for power sources that are consistent and reliable in order to energize the embedded sensors.

Traditional batteries need to be replaced and maintained periodically and hence are not ideal for long-term aerospace applications. Incorporation of piezoelectric material into sandwich structures makes it possible to harness the vibrations from aircraft wings in order to generate electricity and thus create self-powering monitoring systems as well as intelligent aerospace structures[1], [2].

Taking motivation from such innovations, the current study aims to numerically simulate the behaviour of a piezoelectric sandwich panel acting as aircraft wing skin in order to study its structural characteristics as well as its ability to generate power through vibrations[4].

1.6 Problem Statement

Modern aircraft extensively use sensors as well as Structural Health Monitoring system for enhancing safety and effectiveness of maintenance procedures. Still, providing nonstop electricity to the mentioned equipment proves to be a difficult task due to short-term operation of standard batteries and their need in replacements. Meanwhile, constant vibrations during the operation of aircraft wings provide with mechanical energy that is not utilized and is wasted instead[1]. The current research aims to solve the problem in question through studying the application of a piezoelectric sandwich plate within an aircraft wing skin that will allow converting vibrations into electricity[2].

1.7 Objectives

The objectives of the present study are:

1. To develop a sandwich panel representing an aircraft wing skin for vibration-based energy harvesting.
2. To integrate different piezoelectric materials within the sandwich panel.
3. To evaluate the structural behaviour of the proposed configuration using finite element analysis.
4. To perform modal, harmonic response, and coupled-field analyses to determine vibration characteristics and electrical output.
5. To compare the energy harvesting performance of the selected piezoelectric materials.
6. To assess the feasibility of using the harvested energy for self-powered structural health monitoring applications.

1.8 Scope of the Present Work

The current research paper is concerned with the numerical modeling and analysis of a piezoelectric sandwich plate as an aircraft wing skin for energy harvesting through vibrations. In order to conduct the analysis, a three-dimensional finite element model is proposed.

The research involves static structure, modal, harmonic response, and piezoelectric coupled field analyses to analyze deformations, stress distribution, natural frequencies, vibration response, and voltage generation. Static structural, experimental validation, aerodynamic, fatigue, thermal analysis, and fluid-structure interaction are not covered in the current research paper. The results obtained in the current research can form the basis for future research on smart and self-powered aerospace structures[2].

Chapter 2 Literature Review

[2] This study considered a piezoelectric sandwich plate having embedded sensors and actuators layers for vibration control. The mathematical model was based on laminate theory, Hamilton's principle, and Rayleigh-Ritz approach in order to examine the influence of boundary conditions, placement of piezoelectric patches, control voltage, and feedback gains. The findings indicated that the placement of piezoelectric patches near the fixed edge is quite useful for vibration reduction and has an impact on natural frequencies of the plate. This study was confined to the vibration control in a sandwich plate without any consideration of energy harvesting, commercially available PZT materials, or application in an aircraft wing. The current study attempts to overcome such shortcomings.

[6] This study focused on the application of piezoelectric materials for energy harvesting and structural health monitoring (SHM). Various types of piezoelectric structures, sensing methods, and energy harvesting approaches have been addressed in this study, and how piezoelectric transducers integrated in structures harvest electrical energy during structural vibrations. This study demonstrated how using energy harvesting together with SHM could be beneficial for increasing the efficiency of smart structural systems as well as minimizing the use of batteries or any other external energy sources. Nevertheless, this study does not compare various commercial PZT materials for one aerospace structure nor investigate how they affect structural and electrical behavior. This study is intended to fill this gap by comparing five PZT materials in the aircraft wing structure.

[4] In this investigation, the mechanical behavior of the sandwich panel with 3D printed PLA and ABS polymer cores along with 2024-T3 aluminum face plates was studied via experimentation and finite element analysis. It has been observed from the results that the core material and shape play an important role in stiffness, strength, and energy absorption capability of sandwich structure. Moreover, it has been observed from the numerical analysis that the results have good agreement with the experimental findings. But the current investigation does not cover vibration energy harvesting, various piezoelectric materials commercially available, and its application in aircraft wing.

[7] A summary of the latest advances in piezoelectric energy harvester technology was discussed in terms of piezoelectric materials, transducer design, modelling approaches, interface circuits, and engineering applications. In particular, the promise of piezoelectric harvesters as a sustainable source of energy for low power electronics, wireless sensor networks, aerospace systems, and smart structures was stressed, along with the significance of choosing suitable piezoelectric materials and proper structure design to improve the efficiency of energy conversion. Yet, this paper did not include any comparative analysis of commercially available piezoelectric materials with respect to a single structural design, nor does it analyze a sandwich aircraft wing subjected to similar loadings and boundary conditions. The current research intends to fill this gap.

[1] It is worth mentioning that the review has presented the latest trends in the field of piezoelectric energy harvesters concerning the piezoelectric materials used, transducers, modeling methods, electronic circuits, and applications in engineering. It has been mentioned that the progress in materials and devices used in vibration energy harvesters increased their efficiency for aerospace applications, civil structures, and wireless sensors applications. At the same time, this research was aimed at reviewing the latest results but did not include a comparison of different commercial piezoelectric materials in an aerospace structure under the same conditions. The current research includes the comparison of PZT-4, PZT-5A, PZT-5H, PZT-8, and PZT-26 in the sandwich aircraft wing.

[5] The research paper examined a piezoelectric energy harvesting system composed of an auxetic honeycomb sandwich structure featuring face sheets reinforced with CNTs. An analytical electromechanical model with higher order shear deformation theory and Hamilton's principle was formulated to examine the effect of the geometrical core properties, CNT distribution, and mechanical structure properties on the generated voltage and electrical power. The study found that the auxetic core facilitated deformation and lowered the resonance frequency in addition to increasing the energy harvesting capacity of the system. However, the research paper considered a specific type of sandwich panel without comparing other commercial piezoelectric materials and without studying the structure of an aircraft wing. In this study, the work will be expanded through the comparative analysis of five types of PZT materials placed in an aircraft wing.

[8] This study aimed at employing embedded piezoelectric sensors and actuators in the analysis of vibration behavior and structural integrity of composite plates. Through analytical modeling, finite element analysis and experimental verification, it was revealed that piezoelectric sensors can effectively monitor structural behavior changes and damage identification using vibration-based monitoring approach. Nevertheless, the study did not consider vibration energy harvesting application and compared different commercially available piezoelectric materials in a case study of an aircraft wing. This present study attempts to overcome these weaknesses by considering a case study of a sandwich aircraft wing comprising of five types of PZT materials and finite element modeling.

[9] This study has considered the implementation of multiple piezoelectric harvesters in composite sandwich panel structures suitable for aerospace industry. Through finite element analysis, the authors proved that the implementation of piezoelectric harvesters in the composite sandwich panels not only gives protection against environmental effects but also supplies energy to the structural health monitoring systems. Furthermore, it has been recommended in the study that multiple piezoelectric harvesters with different resonant frequencies could be used for broadband energy harvesting in order to maximize the harvested energy. However, the study was conducted keeping in mind the embedded energy harvesting system, and no comparison among different commercial piezoelectric materials or aircraft wing structure has been considered. This study overcomes these shortcomings by comparing five PZT materials implemented in sandwich aircraft wings through finite element analysis.

[3] This research explored the impact of piezoelectric fiber sizes, volume fraction, and matrix material characteristics on the energy harvesting capability of Macro Fiber Composites (MFCs) through a finite element electromechanical analysis. It was found that the output voltage rose with respect to the number of piezoelectric fibers up to a certain optimum value after which the voltage fell because of increasing stiffness and strain inefficiency. The findings indicated that the stiffness of the matrix material and fiber configuration were also key factors in the energy harvesting capability. However, the study had limitations in that it only involved MFCs without comparing the bulk commercial piezoelectric materials in an aerospace context. This research will try to overcome such limitations by exploring the five PZT materials embedded in a sandwich aircraft wing under similar finite element analysis.

[10] This paper reviewed the application of piezoelectric materials for energy generation and looked at the practical ways through which energy harvesting may be done using applications such as piezoelectric tiles, pavements, railways, and breakwaters. Comparison of the energy-generation capacity of these applications and the implications associated with these energy sources was conducted. It was found out that piezoelectric systems can produce electrical energy from the mechanical energy. However, the current study only looked at the urban applications and did not look at the energy harvesting from aerospace structures nor did it compare PZT materials under the same structural condition. This paper aims at comparing five PZT materials used in a sandwich aircraft wing structure.

[11] In the paper, the researchers designed the flexible double-layer piezoelectric nanogenerators using 1-D and 2-D ZnO nanostructures with a Ni-foam interlayer in between them. They conducted their study experimentally where they prepared and examined the single- and double-layer generators subjected to varying mechanical forces and frequency. It was found that the double-layer sandwich type device generated much higher voltage compared to the corresponding single layer device due to its multilayer structure and the presence of intermediate Ni-foam electrode. This paper illustrated that the structural configuration plays a vital role in the enhanced performance of piezoelectric energy harvesters. But it considered only the ZnO based nanogenerators for small scale mechanical energy harvesters and not aerospace structures nor PZT materials which is commercial.

Chapter 3 Theoretical Background

3.1 Introduction

The advent of intelligent aerospace structures has led to the incorporation of smart materials with traditional airplane parts in order to enhance structural properties, along with adding new features like vibration sensing, active control, and energy harvesting. Of all the smart materials, piezoelectric materials have received substantial attention due to their property of conversion between mechanical and electrical energy. The combination of these materials with lightweight sandwich structures results in dual properties of load carrying and energy harvesting, making them ideal for aerospace applications[7].

To successfully develop the proposed piezoelectric sandwich aircraft wing, it is important to have knowledge on some theoretical concepts such as piezoelectricity, piezoelectric materials, sandwich structures, vibration of the aircraft wings, energy harvesting, SHM, and basics of the finite element analysis. All the above-mentioned theoretical concepts make up the scientific basis behind the behaviour of the proposed smart structure[6]. The relationships among these theoretical concepts are illustrated in Figure 3.1, showing the logical sequence of the theoretical concepts adopted in the current study. As seen from the figure, all the theoretical concepts are not treated separately but rather interconnected.

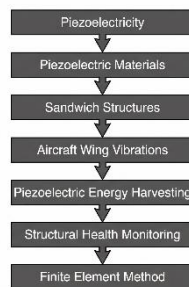


Figure 3.1 : Relationship between the theoretical concepts discussed in this chapter.

Figure 3.1 shows the theory behind which is essential for the interpretation of the current research work. Starting from the basics of piezoelectricity, the chapter sequentially addresses the material and structural aspects needed for vibration energy harvesting of an aircraft wing panel sandwich structure. This is done to lay the engineering foundation for the modeling and simulation of the subsequent chapters[1].

3.2 Fundamentals of Piezoelectricity

3.2.1 Piezoelectric Effect

Piezoelectricity is the process whereby some materials are able to transform mechanical energy into electrical energy, and vice versa. It is because of this property that piezoelectric materials differ from engineering materials and are able to act as transducers. This unique feature has made piezoelectric materials applicable in the aerospace industry, SHM systems, vibration control, and energy harvesting applications. The phenomenon of piezoelectricity was discovered in 1880 by Jacques and Pierre Curie during their study of quartz crystals. They noticed that when mechanical pressure is applied on some crystalline substances, there would be an accumulation of electrical charges on their surface. It was not long after that Gabriel Lippmann theoretically postulated the reverse effect, which was proven to be true by the Curie brothers[10].

Figure 3.2 below shows the two modes of operation of piezoelectric materials, which include the direct and inverse piezoelectric effects along with the polarization process that gives rise to the electromechanical coupling.

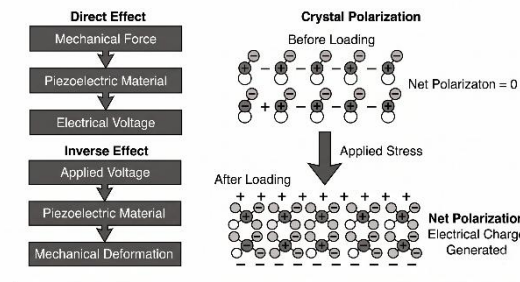


Figure 3.2 : Direct and inverse piezoelectric effects with crystal polarization.

The direct piezoelectric effect changes mechanical loading to electrical energy and works as the basis for piezoelectric sensors and vibration energy harvesters, whereas the inverse piezoelectric effect generates mechanical strain in the presence of an electric field and thus enables the use of piezoelectric materials as precise actuators. The direct and inverse piezoelectric effects result from the same electromechanical coupling process and can happen only in materials having a non-centrosymmetric crystal structure. The practical piezoelectric ceramics like PZT are prepared using the process of poling that aligns the internal ferroelectric domains of the material, which in turn increases the piezoelectric effect[2], [6], [7].

3.2.2 Piezoelectric Constitutive Equations

The behavior of the piezoelectric material is characterized by the use of constitutive relations, which provide an equation linking the mechanical and electrical fields. The constitutive relation explains the generation of electrical displacement due to mechanical stress in piezoelectric material because of the direct piezoelectric effect, and the generation of mechanical strain due to the electric field in piezoelectric material due to the inverse piezoelectric effect. The coupling nature of constitutive relations provides the theoretical basis for analyzing piezoelectric material applications such as sensing, actuation, and energy harvesting based on vibration [10]. Figure 3.3 shows a schematic representation of the interaction between the mechanical and electrical domains.

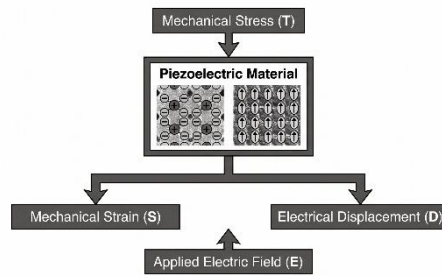


Figure 3.3 : Electromechanical coupling behaviour of a piezoelectric material.

The linear constitutive equations governing piezoelectric behaviour are expressed as follows.

$$S = s^E T + d^T E \quad (3.1)$$

Where,

- S = Mechanical strain
- s^E = Elastic compliance at constant electric field
- T = Applied mechanical stress
- d = Piezoelectric strain coefficient
- E = Applied electric field

(3.1) shows that the total strain generated in a piezoelectric material is due to the combination of mechanical stress and electric field. This phenomenon provides the principle of piezoelectric actuation and structural deformation [10].

$$D = dT + \epsilon^T E \quad (3.2)$$

Where,

- D = Electric displacement
- d = Piezoelectric coefficient
- T = Mechanical stress
- ϵ^T = Dielectric permittivity at constant stress
- E = Electric field

(3.2) shows the electrical behavior produced from the piezoelectric material subjected to mechanical and electrical loads. During vibration energy harvesting, the mechanical stress produces the electrical charge, whereas the dielectric term represents the piezoelectric material's reaction due to the electric field[7]. As seen in the constitutive relations, the electrical and mechanical behaviors of piezoelectric materials are continuously coupled. Hence, both behaviors need to be taken into account when analyzing piezoelectric materials in energy harvesting systems[10].

3.3 Piezoelectric Materials

Piezoelectric materials can be found in several types, each having different electromechanical and mechanical properties. From all the types of piezoelectric materials, the most common are the Lead Zirconate Titanate (PZT) ceramics due to their high electromechanical coupling factor, high energy conversion efficiency, high mechanical stiffness and stability of electrical characteristics. Because of those features, PZT is ideal for sensing and actuating applications, vibration control, and vibration energy harvesting. In aerospace engineering, PZT materials are often included into composite and sandwich structures to transform vibrations of such structure into electrical energy and at the same time to add minimum weight to a structure[7].

Due to the different electromechanical properties of piezoelectric materials that depend on their composition and polarisation process, choosing an appropriate type of material is very important for the engineering application. Generally, soft PZT ceramics have better piezoelectric sensitivity and output electrical power, while hard PZT ceramics have higher mechanical stiffness and stability when subjected to continuous excitation. To present a comparative table of the commercially available PZT ceramics, Table 3.1 shows main properties and applications of the materials.

Table 3.1 : Comparison of commonly used commercial PZT ceramics

Material	Classification	Main Characteristics	Typical Applications
PZT-4	Hard PZT	High stiffness, low dielectric loss	High-power transducers, vibration control
PZT-5A	Soft PZT	High sensitivity, balanced electromechanical properties	Sensors, actuators, energy harvesting
PZT-5H	Soft PZT	Very high piezoelectric coefficient and coupling	Precision actuators, low-frequency energy harvesting
PZT-8	Hard PZT	High mechanical quality factor, excellent durability	Ultrasonic devices, aerospace transducers
PZT-26	Modified PZT	Balanced mechanical and electrical properties	Smart structures and sensing applications

From the above comparison, it is evident that each individual PZT ceramic has unique properties that make it appropriate to be used in various engineering applications. In general, soft PZT ceramics offer higher electric response due to their large piezoelectric constants, while hard PZT ceramics demonstrate higher mechanical robustness and durability. This means that the choice of the appropriate PZT ceramic depends on how well the structure and electrical performance are balanced[3].

From the above comparisons, it can be noted that there is no one material that can perform better in all applications. Higher piezoelectric sensitivity materials generate more electrical energy while stiffer materials have better mechanical strength. Thus, the choice of PZT material will be determined by the performance criteria of the intended application. The PZT commercial materials used in the current study and the reasons for the choice will be highlighted in chapter four[7].

3.4 Sandwich Structures

Sandwich structures refer to lightweight structural configurations where there are two stiff facesheets with lightweight core materials between them. The combination results in increased bending stiffness, good load carrying capacity, and higher stiffness-to-weight ratios when compared to equivalent solid structures of comparable weight. Tensile and compressive stress is mostly carried by the face sheets when there is bending and shear loads are transferred by

the core while keeping the distance between the outer skins, hence improving the structural efficiency. Such properties have resulted in their extensive usage in aerospace engineering[4].

The growing needs for lighter and high-performing aircraft structures have led to increased use of sandwich panels for applications such as aircraft wings, fuselage panels, control surfaces, helicopter blades, spacecraft structures. Their capability to combine the features of lightweight and high stiffness has also rendered them suitable for use in smart structures with piezoelectric materials embedded for sensing and energy harvesting purposes[5], [11]. A smart sandwich structure's elementary architecture is depicted in Figure 3.4, which shows the design of the face sheets, light-weight core, and the piezo-materials that offer structural strength and electromechanical capability.

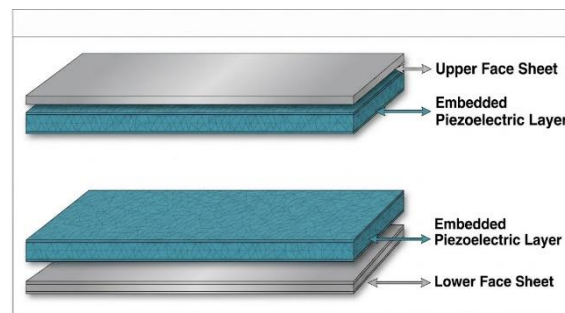


Figure 3.4 : Typical smart sandwich structure with embedded piezoelectric layers.

The layout in Figure 3.4 below shows how each of the individual layers affects the overall performance of the composite. The face sheet layer provides flexural strength while the core layer adds stiffness by providing resistance to shear forces while the piezoelectric layers provide sensors and energy harvesting abilities without adding structural weight. Such a design allows the sandwich structure to carry out both mechanical and electromechanical operations at once. As individual parts play individual roles in the composite, Table 3.2 below summarizes the main functionalities of the layers of the sandwich structure.

Table 3.2 : Functional role of sandwich structure components

Component	Primary Function	Contribution
Face Sheets	Resist tensile and compressive stresses	Provide bending stiffness and structural strength
Core	Transfer shear loads and maintain face sheet separation	Improves stiffness-to-weight ratio while reducing structural mass

Piezoelectric Layers	Convert mechanical strain into electrical energy	Enable sensing, actuation, and vibration energy harvesting
----------------------	--	--

Combining these elements leads to the creation of a light, strong, and multi-purpose structure, which is well-suited to be applied in modern aerospace systems. In contrast to traditional sandwich plates that comprise face sheets and a lightweight core only, the addition of piezoelectric elements creates a smart system, which can carry mechanical loads and generate electricity from structural vibrations simultaneously. These benefits make the sandwich structure a good choice for energy harvesting through vibration and structural health monitoring[6], [9].

3.5 Aircraft Wing Structural Behaviour

The design of aircraft wings takes into consideration the production of lift as well as the ability to withstand various loads that the aircraft experiences while in the air. In terms of structure, the aircraft wings can be described as cantilever beams where the wings are attached to the fuselage at one end while being free at the other end. While in use, the internal structures of the wings such as spars, ribs, and wing skin combine in order to resist various loading effects and maintain the aerodynamic form of the wing. Structural efficiency plays a key role in ensuring high strength in minimal weight which makes this design more favorable in modern aircraft[2]. The aerodynamic and structural loading effect acting on an aircraft wing is depicted in Figure 3.5 below.

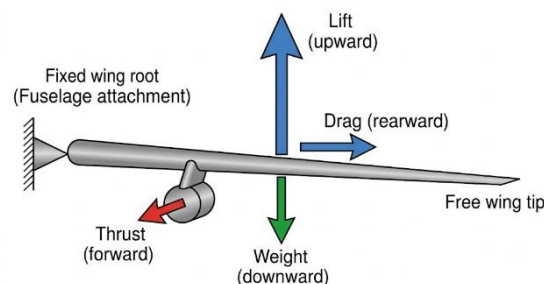


Figure 3.5 : Primary forces acting on an aircraft wing during flight.

The bending moment due to the distributed lift force creates maximum stresses in the wing at its root end since it is there that the wing is mounted on the fuselage. However, the aerodynamic load may also induce torsional moments since the aerodynamic force may not necessarily act through the shear centre. This necessitates the use of spars, ribs, and torsion boxes in the design of aircraft wings to ensure that there is adequate stiffness and stability of the wing[2]. The combined bending and twisting of a cantilever wing are shown in Figure 3.6 below.

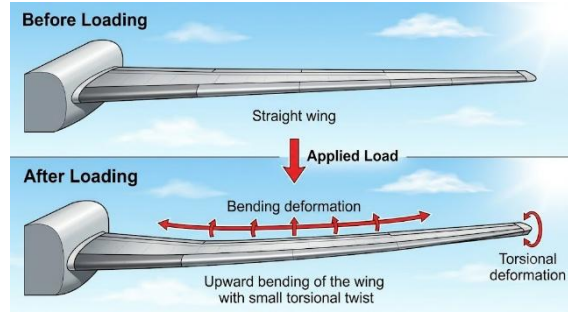


Figure 3.6 : Typical bending and torsional deformation of an aircraft wing

Wing deformation in an airplane is based on several elements including aerodynamic loading, material behavior, structural stiffness, and internal structure of the wing itself. Overbending and twisting could lead to lower performance of the structure and also to some problems related to aeroelastic effects like flutter and divergence. That is why knowledge of aircraft wings structural behavior is vital for design of efficient aerospace structures[2].

3.6 Governing Equations

Mathematically, the theory concepts discussed above can be explained by using some governing equations that explain the structural and electromechanical behaviors of the designed smart aircraft wing. These equations form the backbone of the analysis that enables one to determine stress, strain, elastic deformation, aerodynamics, vibration behavior, and piezoelectric effects. While the current study adopts finite element method for analyzing these equations, it is equally important to note that these equations are useful in explaining the physical behavior of the system[10].

Stress represents the internal resisting force developed within a material when subjected to an external load. It is one of the primary parameters used to evaluate the structural integrity of engineering components.

$$\sigma = \frac{F}{A} \quad (3.3)$$

Where,

- σ = Normal stress (Pa)
- F = Applied force (N)
- A = Cross-sectional area (m²)

The (3.3) shows that the amount of stress induced in the structure depends on the loading and the area through which the load acts. In the current study, the von Mises stress derived from

both the structural and coupled-field harmonic analysis is employed to determine if the aluminium wing is in the elastic region of operation.

Strain represents the relative deformation experienced by a material due to an applied load.

$$\varepsilon = \frac{\Delta L}{L} \quad (3.4)$$

Where,

- ε = Normal strain
- ΔL = Change in length (m)
- L = Original length (m)

The deformation of the material in relation to its original dimensions is given by (3.4). In the numerical studies, the equivalent elastic strain is calculated to ensure that the deformation stays below the elastic limit of the material. Within the elastic region, stress and strain are proportional.

$$\sigma = E\varepsilon \quad (3.5)$$

Where,

- σ = Stress (Pa)
- E = Young's modulus (Pa)
- ε = Strain

The Hooke's Law acts as the foundation for linear elastic analysis in structures and it is suitable for the aluminum alloy material that is used to construct the wing in this case. Finite element simulation takes into consideration Hooke's law in computing the deflection and stress pattern in the wing[10]. The aerodynamic pressure acting on the aircraft wing is related to the dynamic pressure of the airflow.

$$q = \frac{1}{2} \rho V^2 \quad (3.6)$$

Where,

- q = Dynamic pressure (Pa)
- ρ = Air density (kg/m³)
- V = Flight velocity (m/s)

The dynamic pressure is basically the kinetic energy contained within each unit volume of air flow. For this particular study, this principle will be used to establish the representative pressure load acting on the wing during structural analysis.

Despite the fact that the value of pressure used in calculations was chosen to be average, this equation gives an explanation of how to calculate aerodynamic pressure. The natural frequency of a vibrating structure depends upon its stiffness and mass.

$$f_n = \frac{1}{2\pi} \sqrt{\frac{k}{m}} \quad (3.7)$$

Where,

- f_n = Natural frequency (Hz)
- k = Structural stiffness (N/m)
- m = Mass (kg)

This shows in equation (3.7) that an increase in structural stiffness results in an increase in the natural frequency, while an increase in mass leads to a reduction in the same. The explanation for the small difference in modal frequency of the five piezoelectric materials studied is provided by this relation.

The governing equations provided in this section are meant to govern the general mechanical, aerodynamic, and electromechanical behavior of the proposed smart aircraft wing. While the finite element solver provides a numerical solution for these equations, it is through them that a theoretical basis is provided for the understanding of the results from the static structure, modal, and harmonic coupled field analysis.

Chapter 4 Materials Selection and Research Methodology

4.1 Introduction

This chapter outlines the entire methodology used in analyzing the vibration energy harvesting potential of the piezoelectric sandwich panel incorporated into the aircraft wing. Given that the current study is completely computer based, the research employs a methodology comprising of computer aided design (CAD), finite element modeling, structural analysis and performance evaluation. The methodology outlines a systematic process through which the current model can effectively capture the behavior of the proposed aircraft wing.

It is important to note that this chapter does not contain any theoretical principles or simulation results; instead, it outlines the entire process involved in the project. Specific modeling processes, simulation parameters and results will be provided in the next chapters.

4.2 Research Methodology

In the current study, the methodology for conducting the simulation study was chosen in order to analyze the structure and energy harvesting capabilities of the smart wing for aircrafts, made of piezoelectric material. The use of a numerical method is preferred as it allows to examine the behavior of the structure and its electromechanical properties under controlled loads prior to physical experiments[8].

Initially, a thorough examination of literature was conducted to gain information on the latest advancements in piezoelectric energy harvesting, sandwich structures, design of aircraft wings and structural health monitoring. Based on the data obtained, the appropriate materials and the whole wing structure were chosen. The entire research methodology used in this study is shown in Figure 4.1 below.

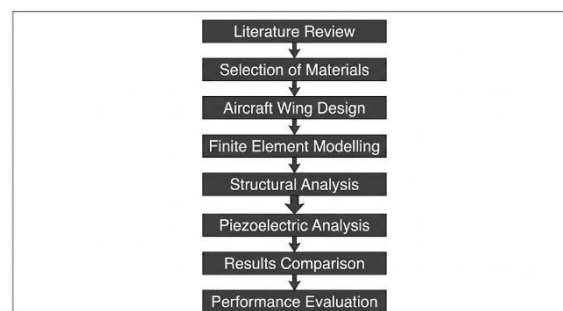


Figure 4.1 : Overall research methodology adopted for the present investigation.

The methodology involved the use of a systematic process where the selected materials as well as the configuration of the aircraft wings were analyzed numerically, and their responses to

identical conditions were compared. This helps in providing a consistent means of analyzing their structural behavior and their ability to harvest energy while developing smart aerospace structures[8].

4.3 Software Tools Used

The present work was completed using commercially available engineering software packages that are widely employed in aerospace design and finite element analysis.

Table 4.1 : Software used during the project

Software	Purpose
CATIA V5	Three-dimensional modelling of the aircraft wing
ANSYS Workbench	Finite element modelling and structural analyses

Each software package was selected according to its capability to perform specific tasks within the overall simulation workflow. CATIA V5 was employed for geometric modelling, whereas ANSYS Workbench was used to evaluate the structural behaviour of the proposed smart wing under different loading conditions.

4.4 Parameters Evaluated

Engineering response parameters for the assessment of the structural and electromechanical properties of the designed intelligent aircraft wing have been investigated. The parameters chosen are those to be used in the assessment of the mechanical behavior, dynamics, and energy harvesting potential of the sandwich wing under the specified loadings. The engineering response parameters considered in this study are shown in Table 4.2.

Table 4.2 : Parameters evaluated during the numerical investigation

Parameter	Significance
Total Deformation	Determines the overall structural displacement under applied loading.
Equivalent Stress (von Mises)	Evaluates the structural strength and identifies regions of maximum stress concentration.
Equivalent Elastic Strain	Assesses the elastic deformation developed within the wing structure.

Natural Frequency	Determines the dynamic characteristics and resonance behaviour of the aircraft wing.
Output Voltage	Measures the electrical potential generated by the embedded piezoelectric material due to structural vibration.
Electrical Power	Evaluates the energy harvesting capability of the selected piezoelectric materials.

Parameters listed in Table 4.2 are considered as a full range of criteria to evaluate structural soundness and electro-mechanical properties of the proposed sandwich wing. Mechanical characteristics will be measured through deformation, stress, and strain, while dynamic properties will be estimated through natural frequency. Effectiveness of vibration-based energy harvesting will be evaluated based on the analysis of the generated voltage and power through piezoelectric sensors. These results will be discussed in [Chapter 7](#)[8].

4.5 Assumptions and Limitations

In order to make the numeric study simpler and consistent with all finite element analysis conducted for the current study, the following assumptions were used in this case:

- The wing of the aircraft was taken to be in the linear elastic region, where all the materials were not going to reach the yielding point[2].
- The bonding between the aluminum face sheets, core and piezoelectric layers was taken to be perfect with no debonding taking place[2].
- The aluminum alloy, core material, and piezoelectric ceramics were homogenous and isotropic in nature while piezoelectric behavior was defined using the material constants provided by the manufacturer[2].
- The loads and boundaries used for analysis are considered to be consistent with the operational conditions of the aircraft wing.
- The environmental factors like temperature variation, moisture content, aging of materials and defects in manufacturing have not been accounted for during numerical analysis.
- Aerodynamic fluid flow analysis, aero-elastic interactions, fatigue behavior, impact loads and damage propagation have not been considered during the current study.

Chapter 5 CAD Modelling

5.1 Introduction

A CAD model of the geometry in three dimensions is required in order to proceed numerically with the analysis of aerospace structures. In this research project, the aircraft wing is modelled using CATIA V5, in order to obtain a precise representation for the subsequent finite element analysis. The parametric modelling technique is employed in order to attain dimensionally consistent geometry which can be easily modified. The new proposed model is a half-wing with rectangular shape using the Clark-Y airfoil section. The wing structure contains two hollow square spars along with six evenly distributed ribs and the skin of the wing is made from a sandwich structure consisting of aluminium plates along with piezoelectric layer inside the sandwich structure. The combination of the wing with both the external aerodynamic shape and internal structure is an example of vibration energy harvesting. The segmented approach to the design of the wing also helps with geometric organization and with the creation of the finite element model.

In this chapter the design process of the wing of the aircraft is described including wing shape, internal structure of the wing, sandwich skin construction, introduction of the piezoelectric layers and the final assembly. The finished CAD model will be used in the further analysis in the next chapter.

5.2 Wing Geometry

In the current research work, the wing design was made such that the wing can act as a representative structure model for vibration-based energy harvesting. A half-wing structure has been chosen in order to represent the full structure as a cantilever structure but with less computational load on numerical analysis. The wing has been designed using the Clark-Y airfoil shape, which will give an appropriate depth to the structure along with the capability to keep sufficient space inside the structure.

This wing has been built to have a fixed chord, thus the wing will have a rectangular shape, making the structure easy to understand and making it easy to create the finite element pre-processing. The internal structure of this wing is made up of two hollow square spars and six equally distributed ribs, and the wing skin is made up of segmented sandwich construction, with aluminum face sheets and piezoelectric layers incorporated in it. This makes it stiff and

easy to manufacture as well as making it easier to create a numerical model. The main geometric specifications used for this wing are shown below in Table 5.1.

Table 5.1 : Wing Geometrical Specifications

Parameter	Value
Wing Configuration	Rectangular Half-Wing
Airfoil	Clark-Y
Half Span	500 mm
Root Chord	115.84 mm
Tip Chord	115.84 mm
Maximum Airfoil Thickness	13.55 mm (11.7% Chord)
Upper Skin Thickness	1 mm
Intermediate Layer Thickness	1 mm
Lower Skin Thickness	1 mm
Total Sandwich Thickness	3 mm
CAD Software	CATIA V5

The entire three-dimensional geometry, developed through CATIA V5, incorporates the aerodynamic shape along with the structural framework that needs to be designed for the envisaged smart wing. This entire structure has been depicted in Figure 5.1 below, where the geometrical structure of the rectangular half-wing has been depicted.

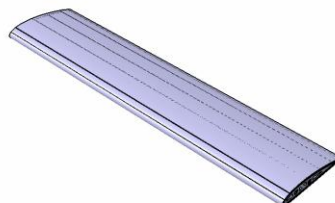


Figure 5.1 : Complete CAD model of the rectangular half-wing developed in CATIA V5.

The CAD model ensures that the external aerodynamics shape is maintained while integrating the spars, ribs, and segmented skin into the model. The segmentation of the aluminium face sheets and the piezoelectric material make the manufacturing process easier and also allows

for effective finite element modeling without changing the basic structure. The design of the individual wing components is discussed below.

5.3 Structural Design of the Wing

The internal structure of the proposed aircraft wing is structured in such a way that it ensures enough stiffness and strength while remaining light in weight for vibration energy harvesting purposes. It includes two longitudinal spars along with six ribs that are equally spaced to maintain the shape of the sandwich wing skin along with maintaining the shape of the Clark-Y airfoil. The combination of all the above parts makes up the load path of the wing. The spars are the primary load carrying components of the wing and transfer most of the bending loads developed during its operation. In order to strike a compromise between the strength and weight of the spars, two hollow spars of square cross-sections having an outer size of $5 \text{ mm} \times 5 \text{ mm}$ with a wall thickness of 0.5 mm have been introduced. The distance of the foremost spar from the leading edge is 25.482 mm whereas the distance of the rear spar from the leading edge is 50.481 mm . The spars run continuously over the half span length to provide support to the ribs and sandwich skin. The main characteristics of the spars are listed in Table 5.2 below.

Table 5.2 : Spar Dimensions

Parameter	Value
Number of Spars	2
Cross Section	Hollow Square
Outer Dimensions	$5 \text{ mm} \times 5 \text{ mm}$
Wall Thickness	0.5 mm
Length	500 mm
Front Spar Position	25.482 mm from Leading Edge
Rear Spar Position	50.481 mm from Leading Edge

It is shown from Figure 5.2 how the relationship of the spars and ribs exists before the sandwich skin assembly. This structure design has been achieved to ensure that the stability of the airfoil profile is maintained while having sufficient support for the segmented sandwich.

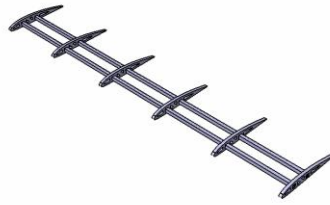


Figure 5.2 : Internal structural arrangement of the wing showing the spars and ribs.

This structural configuration is indicative of how the spars run along the wing span, while ribs keep the airfoil shape and transmit forces between wing skin and spars. This ensures a solid basis for the segmented sandwich skin and overall structural rigidity of the wing. In order to preserve the airfoil shape, there have been six ribs installed with a pitch of 99.4 mm. Every rib is built on the lines of the Clark-Y airfoil shape with lightening holes for weight saving, yet retaining enough rigidity. There is also square spar pockets used to guarantee correct placement of the spars. The main parameters of ribs are given in Table 5.3.



Figure 5.3 : CAD model of the rib showing the lightening holes and spar pockets

Table 5.3 : Rib Dimensions

Parameter	Value
Number of Ribs	6
Rib Thickness	3 mm
Rib Spacing	Approximately 99.4 mm
Airfoil Profile	Clark-Y
Lightening Holes	Provided
Spar Pockets	Two Square Pockets

The joining of two longitudinal spars with six light ribs results in a sturdy internal structure that is able to sustain the external sandwich skin without losing the aerodynamic shape of the

wing. The mentioned construction also allows ample room for the use of the segmented sandwich approach presented below.

5.4 Sandwich Panel Design

The wing skin was constructed with a segmented three-layer sandwich construction in order to offer a combination of structural integrity and vibration-based energy harvesting functionality. This is made up of an upper aluminum face sheet, a segmented middle layer, and a lower aluminum face sheet, all with a thickness of 1mm, hence making the overall sandwich construction thickness 3mm. This allows for the maintenance of the aerodynamic profile on the outer side while ensuring that the structure can incorporate piezoelectric materials[6], [9]. The top and bottom aluminum face sheets were divided into two panels in order to make the assembly simpler while at the same time retaining a continuous outer surface through bonding.

The exploded diagram demonstrates the multi-layered approach used in the construction of the wing skin, as well as the process of assembling the different parts. The segmented face panels offer structural rigidity, yet ensure that the intermediate panel can be inserted without changing the external shape of the wing. The intermediate panel has been split up into eight segments running across the chord direction, consisting of two end segments made from aluminium, and six central segments of piezoelectric (PZT) material. This configuration ensures that all the piezoelectric material is placed in the region of the wing susceptible to vibrations.

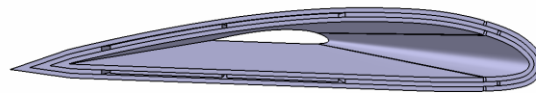


Figure 5.4 : Segmented aluminium and piezoelectric intermediate layer arrangement.

Figure 5.4 provides the explanation for how the aluminium segments contribute to the structural strength whereas the piezoelectric segments perform the role of energy harvesting. Such an arrangement allows for the effective use of piezoelectric elements without compromising the structural strength of the wing. The principal dimensions of the sandwich structure are summarised in Table 5.4.

Table 5.4 : Sandwich Panel Dimensions

Component	Specification
Upper Skin Thickness	1 mm
Intermediate Layer Thickness	1 mm
Lower Skin Thickness	1 mm
Total Sandwich Thickness	3 mm
Upper Aluminium Panels	2
Lower Aluminium Panels	2
Intermediate Layer Segments	8
Aluminium Segments	2
Piezoelectric Segments	6
Intermediate Layer Orientation	Along the Chord

Segmented sandwich concept incorporates lightweight design along with multilayered functions by embedding piezoelectric elements into the structure of the wing skin. The selected design configuration allows carrying structural loads and provides a viable basis for vibration energy harvesting process[6], [9].

5.5 Integration of the Piezoelectric Layer

The efficiency of vibration energy harvesting system is determined by how the piezoelectric material is positioned in the structure. The intermediate layer of the sandwich structure was made to have a segmented design in order to enhance strain harvesting without compromising structural integrity. Rather than having the whole layer consisting of a piezoelectric material, it comprises eight chordwise segments where six of them consist of piezoelectric materials (PZT) while two are aluminium end segments. In this way, active material is concentrated on the zone that will harvest vibrations[6], [9].

From the cross-sectional view, the piezoelectric components are fully enclosed by the aluminium facings, ensuring that the piezoelectric components are protected from any external damage and enabling the effective transfer of mechanical stress from the wings. This arrangement improves the electromechanical coupling and is extensively used in smart

sandwich structures for sensing and vibration energy harvesting. The principal characteristics of the embedded piezoelectric layer are summarised in Table 5.5.

Table 5.5 : Configuration of the Embedded Piezoelectric Layer

Parameter	Value
Piezoelectric Material	PZT (Material varied during analysis)
Number of Piezoelectric Segments	6
Number of Aluminium Segments	2
Total Intermediate Segments	8
Segment Arrangement	Along the Chord
Integration Method	Embedded within Intermediate Layer
Active Region	Central Portion of the Sandwich Panel

The piezoelectric structure chosen offers a compromise between mechanical stability and energy harvesting ability through incorporating the active material in the sandwich structure while retaining the aluminium support on either side. The design facilitates effective strain transmission into the piezoelectric elements and serves as the basis for the electromechanical analysis discussed in the following sections.

Chapter 6 Finite Element Analysis

6.1 Introduction

Structural and electromechanical behavior of the suggested smart aircraft wing was analyzed through the Finite Element Method (FEM) in ANSYS Mechanical 2025 R2. FEM is a numerical method used for predicting the response of an engineered structure based on dividing the structure into a finite number of elements and solving the equation based on the defined loading and boundary conditions. This method allows for evaluating the deformation, stress state, vibrations and piezoelectric behavior without the construction of the physical prototype.

Three-dimensional CAD model created in CATIA V5 was imported to ANSYS Mechanical software, and material properties, contacts, meshing and boundary conditions were defined in order to create numerical model. During all the tests, geometry of the wing, the load, and other parameters of the analysis were kept unchanged; the only thing that was varied during the process was the type of piezoelectric material applied to the intermediate layer. The whole scheme of numerical modeling of the structure is shown in Figure 6.1 below.

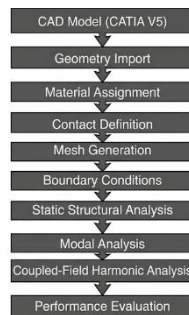


Figure 6.1 : Overall numerical simulation workflow adopted for the present study.

6.2 Engineering Data and Material Properties

The properties of the material needed for the finite element analyses were specified in the Engineering Data section of ANSYS Mechanical prior to the construction of the model. Here the mechanical and electromechanical properties of the materials composing the structure and the piezoelectricity were attributed. Engineering Data is a module that enables the definition of material properties. The aluminium alloy was attributed to the wing skins, spars, ribs and to the aluminium segments of the intermediate layer. The same structural material was kept for all simulations to maintain the same structural behaviour during the whole investigation. The most important mechanical properties of the aluminium alloy used in the numerical model are presented in Table 6.1.

Table 6.1 : Engineering Properties of Aluminium Alloy

Property	Value
Density	2.77e-006 kg mm ⁻³
Young's Modulus	71000 MPa
Poisson's Ratio	0.33
Yield Strength	280 MPa
Ultimate Tensile Strength	310 MPa

The embedded piezoelectric elements were simulated using five different commercial PZT materials. In the comparison, the only aspect that varied was the piezoelectric material used in the intermediate layer while all other aspects including geometry, mesh, contacts, loads, and boundary conditions were kept the same. This helped ensure that the only reason for variation in the simulation results was the material property of the PZT ceramic used.

The Engineering Data environment used to assign the structural and piezoelectric materials is shown in Figure 6.2, illustrating the material definitions employed during model preparation.

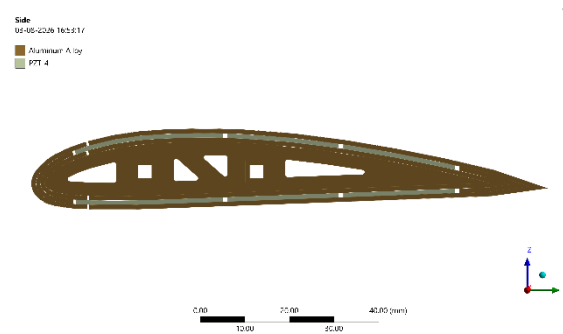


Figure 6.2 : ANSYS Engineering Data environment showing material definitions..

From the Engineering Data configuration given in Figure 6.2, it is clear that the aluminum alloy did not change during the experiment while the five types of PZT material were given individual assignments for comparison purposes. The material definitions set in this phase formed the basis for further FEM simulations.

6.3 Finite Element Mesh Generation

Geometry of the wing of the aircraft was then discretized into finite elements via the meshing tool within the ANSYS Mechanical. An adequate mesh was created that would be able to

capture the structure of the sandwich skins, spars, ribs, and piezoelectric sections, and at the same time, it would allow for solution accuracy and efficiency. Meshing is an important process in the process of finite element modeling. Prior to creating the mesh, it was necessary to set up the global mesh controls which would ensure consistency in element distribution within the model. Finite element mesh obtained is shown in Figure 6.3.

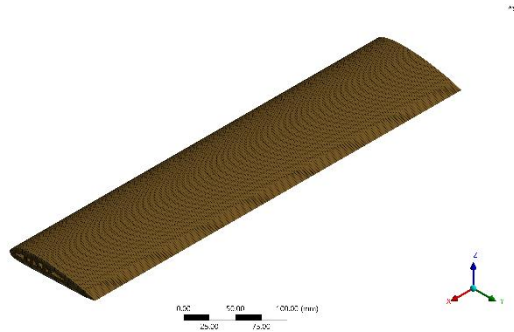


Figure 6.3 : Finite element mesh of the smart wing model in ANSYS Mechanical..

Figure 6.3 reveals that meshing was performed successfully for the wing structure considering all its elements – the sandwich skins, spars, ribs and piezoelectric sections, which ensured continuity between all neighbouring parts of the model. The chosen meshing was kept unchanged in all computations in order to provide uniformity in comparison of various piezoelectric materials. The main features of the mesh used for numerical studies are presented in Table 6.2.

Table 6.2 : Finite Element Mesh Statistics

Parameter	Value
Number of Nodes	608002
Number of Elements	85091

The mesh characteristics outlined in Table 6.3 can give an indication of the type of discretization carried out for the numerical analysis. The mesh was kept constant for the duration of the study to ensure that the observed effects were due solely to variations in piezoelectric properties. This created mesh was then used in the static structural, modal, and coupled field harmonic analyses outlined in the following sections. Contact Definitions and Boundary Conditions

6.3.1 Contact Definitions

Interaction between the structural parts of the airplane wing was modeled by means of Bonded contact in ANSYS Mechanical software package. Bonded contact ensures that the adjacent surfaces do not slide or separate from one another, thus enabling these connected parts to operate as a unified body. This approach is justified by the fact that all the aluminum skins, spars, ribs, and piezoelectric elements in the wing will be rigidly fastened to each other. Moreover, bonded contact can be used in linear analysis, as the state of the contact surface does not change during the load application. Contact conditions were kept the same across all the simulations in order to provide a common basis of comparison between the piezoelectric material types. The contact interfaces used in the numerical simulation are summarized in Table 6.3.

Table 6.3 : Contact Definitions Used in the Numerical Model

Contact Interface	Contact Type
Aluminium Skin – Rib	Bonded
Aluminium Skin – Spar	Bonded
Aluminium Skin – Piezoelectric Segment	Bonded
Piezoelectric Segment – Aluminium Segment	Bonded
Internal Structural Interfaces	Bonded

6.4 Static Structural Analysis

6.4.1 Analysis Setup

The static structural analysis of the proposed smart wing of an aircraft was done using ANSYS Mechanical 2025 R2 software for studying the structural behavior of the proposed smart wing under steady aerodynamic loading. Static structural analysis is done to study the deformations, stresses, and strains induced in the structure when the loads are steady and inertia and damping effects are assumed to be insignificant. The finite element solution is based on the global equilibrium equation:

$$[K]\{u\} = \{F\} \quad (6.1)$$

Where,

- K = Global stiffness matrix

- u = Nodal displacement vector
- F = Applied external load vector

The imported model for finite element analysis used the same geometric shape, material distribution, mesh, bonded contact, and boundary condition as those used in the pre-processing stage. Throughout the comparative analysis process, the only thing that was changed was the piezoelectric material of the middle layer; everything else was kept constant in order to maintain consistency. From the equation (3.6) the load due to aerodynamics was considered as a uniformly distributed pressure on the lower surface of the wing. The value of the pressure was found using the dynamic pressure equation.

$$q = \frac{1}{2} \times 1.225 \times (20.203)^2$$

$$q = 250 \text{ Pa}$$

The calculated pressure was applied uniformly to the lower surface of the wing, while a Fixed Support was assigned at the wing root to represent its attachment to the aircraft fuselage. The complete loading arrangement is illustrated in Figure 6.4.

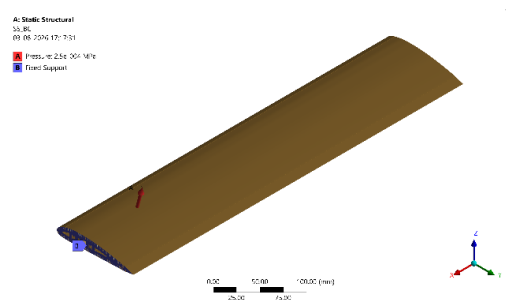


Figure 6.4 : Static structural loading and boundary conditions of the aircraft wing..

Boundary conditions depicted in Figure 6.4 were kept the same for all the simulations so that all the structures were analysed under the same loading conditions. Analysis parameters are briefly presented in Table 6.4.

Table 6.4 : Static structural analysis settings

Parameter	Value
Analysis Type	Static Structural
Solver	ANSYS Mechanical 2025 R2
Applied Pressure	2.5e-004 MPa

Pressure Location	Lower Wing Surface
Support Condition	Fixed Support at Wing Root
Large Deflection	Off
Solver Type	Program Controlled

The parameters in Table 6.4 used in the analysis were held at the same value during the entire study to maintain consistency when comparing the various piezoelectric materials. Structural responses from the analysis will be discussed in [Chapter 7](#).

6.5 Modal Analysis

6.5.1 Analysis Setup

Modal analysis is carried out using ANSYS Mechanical 2025 R2 software to obtain natural frequencies and mode shapes of the proposed smart wing of an aircraft. Modal analysis helps to determine the vibratory characteristics of the structure in its free vibration and forms the basis of harmonic response analysis. Natural frequency determination is critical since resonance can take place at the frequencies equal to those of excitation causing vibration. The governing equation for an undamped free vibration system is expressed as:

$$[M]\{\ddot{u}\} + [K]\{u\} = 0 \quad (6.2)$$

Where,

- M = Global mass matrix
- K = Global stiffness matrix
- u = Nodal displacement vector

Assuming harmonic motion, the equation is transformed into the eigenvalue problem:

$$[K - \omega^2 M]\phi = 0 \quad (6.3)$$

Where,

- ω = Natural circular frequency
- ϕ = Mode shape vector

Solving this equation allows obtaining the natural frequencies and mode shapes of the structure, which mostly depend on its shape, stiffness, mass distribution, and boundary conditions.

In the case of modal analysis, the model was created with the help of the same geometry, material properties, meshing, contact definitions, and support conditions that were created in the pre-processing step. Contrary to the static structural analysis, no loading is applied to the structure, and thus it allows determining the dynamic properties of the wing at the free vibration state. The cantilever condition of the wing was preserved, meaning that the fixed constraint of the wing root was kept.

To describe the dynamic response of the wing, the first six natural vibration modes were identified. This number of modes is enough to reveal the main bending and torsion properties of the structure, and these modes were used as excitation frequencies for the harmonic analysis of the coupled fields. Since all other parameters were kept constant, any difference observed in the natural frequencies obtained was a result of the choice of the piezoelectric material. The main settings used for the modal analysis are listed in Table 6.5.

Table 6.5 : Modal analysis settings

Parameter	Value
Analysis Type	Modal
Number of Modes Extracted	6
Geometry	Same as Static Structural
Mesh	Same as Static Structural
Contact Definition	Bonded
Boundary Condition	Fixed Support at Wing Root
External Mechanical Load	None
Solver	Program Controlled

6.6 Coupled-Field Harmonic Analysis

6.6.1 Analysis Setup

Harmonic analysis of coupled field was conducted in ANSYS Mechanical 2025 R2 to study the coupled behavior of the proposed smart wing of an aircraft. Unlike traditional harmonic analysis, harmonic analysis of coupled fields considers electromechanical coupling effect of the piezoelectric material; hence, it is possible to predict both the mechanical behavior and the induced voltage in one solution. This kind of analysis is appropriate for modeling vibration-

based energy harvesting devices containing piezoelectric materials. The electromechanical behaviour of the piezoelectric material is governed by the constitutive equations (3.1) and (3.2). These equations describe the coupling between the mechanical and electrical fields, allowing mechanical vibration of the piezoelectric material to generate an electrical response. The harmonic response of the structure is governed by:

$$[M]\ddot{u} + [C]\dot{u} + [K]u = F(t) \quad (6.4)$$

where:

- M = Mass matrix
- C = Damping matrix
- K = Stiffness matrix
- u = Displacement vector
- $F(t)$ = Harmonic excitation force

The coupled-field harmonic analysis was conducted with the same geometry, material properties, mesh, bonded contacts, and boundary conditions that had been determined during the pre-processing phase. The range of the excitation frequency was taken from 0 Hz to just higher than the sixth natural frequency, which had been determined from the modal analysis in each case of piezoelectric material. In addition to this, the six natural frequencies that were determined during the process of modal analysis were defined as user defined frequencies with five substeps between them. In this way, the harmonic analysis would study the performance of the structure/electrical properties of the structure within the most significant frequency range.

The electrical boundary conditions were applied on the embedded piezoelectric parts by using the Voltage constraint on the upper electrode of the piezoelectric part and the Voltage (ground) constraint on the opposite electrode. These electrical boundary conditions helped the coupled-field solver to find out the voltage produced by the piezoelectric material with harmonic excitation. The whole process of the coupled-field harmonic analysis used in the current research is presented in Figure 6.5.

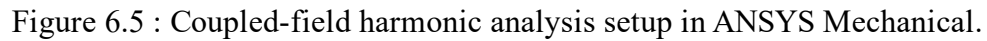


Table 6.6 : Coupled-field harmonic analysis settings

The analysis options shown in Table 6.6 were kept constant for all cases to make a consistent comparison of the chosen piezoelectric materials. The results of the coupled harmonic field analysis in terms of mechanical and electrical behaviors will be discussed in [Chapter 7](#).

Chapter 7 Results And Discussion

7.1 Introduction

In this chapter, the numerical results generated by the finite element analyses of the designed smart wing of the aircraft will be presented and discussed. The purpose of the numerical study was to assess the performance of the smart wing in terms of its structure and electromechanical behavior when made up of various piezoelectric materials. Unlike in the previous chapter where the methodology of the simulation is discussed, the current chapter will discuss the performance of the designed smart aircraft wing using different piezoelectric materials. It is important that results be presented and analyzed in a systematic way. It must be noted that in the numerical study carried out in order to compare the performance of various piezoelectric materials, all parameters such as geometry, meshing, material properties of the aluminum parts, contacts, boundary conditions, and solver settings were fixed. Only the piezoelectric material used for the embedded layer was varied.

Results are arranged in the same order as that of the numerical analysis carried out. In the beginning, results related to the static structural analysis of the aircraft wing are presented. Here, total deformation, equivalent (Von Mises) stress and equivalent elastic strain are analyzed. Next comes the mode analysis, which involves finding the natural frequencies and mode shapes. The last part of the chapter will deal with the coupled field harmonic analysis to determine the voltage generation and energy harvesting performance of the embedded piezoelectric materials. For the purpose of better understanding of the results, they are presented through contour plots, comparison tables and graphs. In this way, a comparison between the properties of PZT-4, PZT-5A, PZT-5H, PZT-8 and PZT-26 can be made. This helps to identify the appropriate material for the proposed smart sandwich aircraft wing. Contour plots, graphs and comparison tables should be effectively processed and arranged to interpret the finite element results.

7.2 Static Structural Analysis Results

7.2.1 Total Deformation

The total deformation is an important parameter that is used to study the performance and stiffness of the smart wing subjected to aerodynamic loading. Due to the fact that the geometry, meshing, boundary conditions, load, and other parameters were kept same in all the simulation models, the difference in maximum deformation is due only to the mechanical properties of the

piezoelectric material. The maximum total deformation for each piezoelectric material is presented in Table 7.1.

Table 7.1 : Maximum total deformation for different piezoelectric materials.

Piezoelectric Material	Maximum Total Deformation (mm)
PZT-4	0.1459
PZT-5A	0.1583
PZT-5H	0.1588
PZT-8	0.1427
PZT-26	0.1387

The results of comparison given in Table 7.1 demonstrate that there is some difference in the maximal deformation values between the examined piezoelectric materials. The minimal value of deformation equals to 0.1387 mm for PZT-26, and the maximal value of deformation equals to 0.1588 mm for PZT-5H. The observed discrepancies suggest that the stiffness of the incorporated piezoelectric material does not significantly affect the rigidity of the entire structure of the smart wing. The respective deformation contours are provided in Figure 7.1.

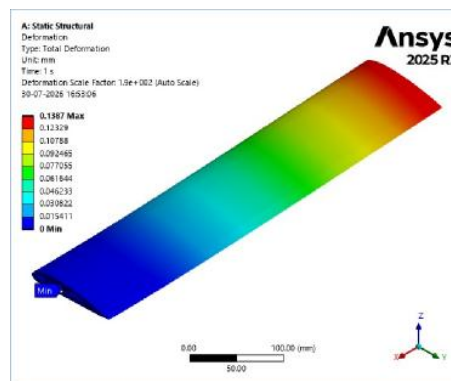


Figure 7.1 : Total deformation contour of PZT - 26.

All five piezoelectric materials in Figure 7.1 show the same deformation behaviour; that is, they have minimum deformation at the fixed wing root and maximum at the free wing tip, which is typical for a cantilever wing subjected to distributed loading. While all materials have a common deformation behaviour, there are minor differences in their maximum deformations due to their different mechanical properties. This means that the use of the piezoelectric material affects the stiffness of the smart wing without changing the deformation behaviour of the wing. The greater stiffness of a structure is seen from smaller deformation under the same

load conditions. But deformation is not enough to find out the optimum material since the overall properties are dependent on stress, strain, dynamic properties and electrical energy harvesting ability. All these properties are considered further in this chapter.

The total deformation contour plot for PZT-26 is illustrated in Figure 7.1. The complete series of static total deformation contour plots for all five PZT materials are included in [Appendix B](#).

7.2.2 Equivalent (von Mises) Stress

Equivalent (von Mises) stress was calculated to determine the structural behavior of the intelligent airplane wing subjected to the aerodynamic loads. As the same geometry, mesh, loads, boundaries, and contacts were used in all cases, any difference in the maximum von Mises stress can be attributed solely to the mechanical properties of the piezoelectric material present in the wing. Maximum von Mises stress results for all materials are tabulated in Table 7.2, which gives the maximum stress in the wing due to the applied static loads.

Table 7.2 : Maximum equivalent stress for different piezoelectric materials.

Piezoelectric Material	Maximum Equivalent Stress (MPa)
PZT-4	1.8991
PZT-5A	2.0372
PZT-5H	2.0324
PZT-8	1.849
PZT-26	1.8923

As seen from the comparison made in Table 7.2, there is little difference in the value of the maximum equivalent stress in the analyzed piezoelectric materials. The minimum stress value of 1.849 MPa was found for PZT-8, while the maximum stress value of 2.0372 MPa was determined for PZT-5A. This difference arises due to the difference in mechanical properties of the embedded piezoelectric materials. The stress contour plots are given in Figure 7.2.

Figure 7.2 below shows the equivalent von Mises stress contour plot for PZT-26. Equivalent stress contour plots for all five PZT materials are presented in [Appendix B](#).

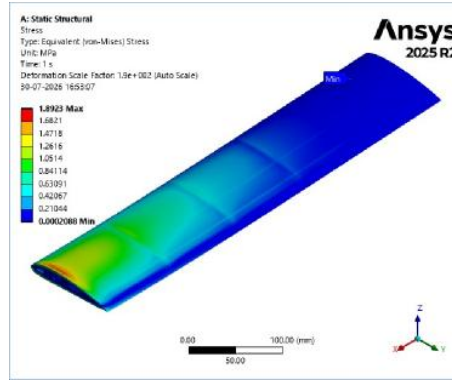


Figure 7.2 : Equivalent stress contour of PZT - 26.

It can be seen from Figure 7.2 that the stress distribution is more or less same for all the five piezoelectric materials with the highest stresses being present at the fixed wing root due to maximum bending moment and decreasing gradually to the wing tip. Though there are some small variations in the maximum stresses, however, the distribution of stress still remains same. The highest value of the equivalent stress is well below the yield strength of aluminium alloy (280 MPa) showing that the wing lies within the elastic range and is safe structurally for all the material combinations. From the engineering point of view, the lower values of equivalent stress mean more structural safety factor. But only the stress can't be taken into consideration while finding the best piezoelectric material.

7.2.3 Equivalent Elastic Strain

Elastic strain equivalency was investigated for determining the elastic deformation characteristics of the smart aircraft wing when subjected to the applied aerodynamic loading. As all simulation studies were conducted using the same conditions, any difference in the elastic strain equivalency must be due to the mechanical characteristics of the incorporated piezoelectric material. Elastic strain equivalency calculated using ANSYS is based on the strain tensor in the von Mises sense and is proportional to the stress distribution. Table 7.3 below shows the maximum equivalent elastic strain for each piezoelectric material.

Table 7.3 : Maximum equivalent elastic strain for different piezoelectric materials.

Piezoelectric Material	Maximum Equivalent Elastic Strain (mm/mm)
PZT-4	2.9862e-005
PZT-5A	3.2522e-005
PZT-5H	3.3532e-005

PZT-8	2.8339e-005
PZT-26	2.7441e-005

The comparison shown in Table 7.3 reveals that there are minor differences in the value of the maximum equivalent elastic strain in the investigated piezoelectric materials. The minimum strain in the range of 2.7441e-005 occurred in PZT-26, whereas the maximum strain of 3.3532e-005 occurred in PZT-5H. The cause of these differences lies in the difference in the mechanical properties of the piezoelectric materials embedded in the structure. The contour maps of the strains in question are shown in Figure 7.3. The equivalent elastic strain calculated by ANSYS is based on the strain field in the von Mises sense.

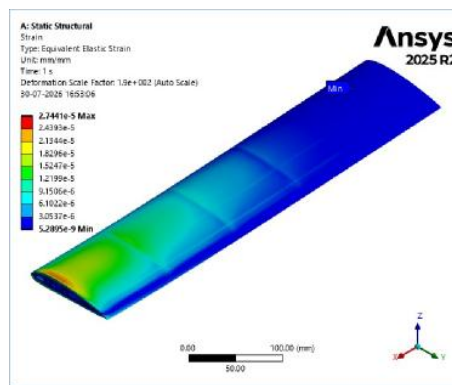


Figure 7.3 : Equivalent elastic strain contour of PZT - 26.

As illustrated in Figure 7.3, the strain distribution corresponds well with the stress distribution, whereby the largest strains are localized close to the wing root, and the strains gradually diminish along the wing span up to the free wing tip due to the distributed loading of a cantilever beam. The relatively small maximum value of the equivalent elastic strain suggests that the wing is subject to pure elastic deformations, which confirms sufficient stiffness of the wing under all studied material combinations. The equivalent elastic strain value outputted by ANSYS software is related to the stress field, and it is evaluated by the von Mises formula.

Figure 7.3 displays the contour plot of equivalent elastic strain for PZT-26. The contour plot of equivalent elastic strain for all five PZT materials can be found in [Appendix B](#).

From an engineering point of view, the smaller elastic strains mean higher structural stiffness, i.e., less deformation under identical loads. But the difference between the tested materials is not high since the stiffness of the wing is mainly determined by the aluminium skin, spars, and ribs.

7.3 Modal Analysis Results

7.3.1 Natural Frequency Analysis

Modal analysis was carried out in order to compute the first six natural frequencies and mode shapes of the developed smart wing of an aircraft. The modal analysis will give the frequencies at which resonance could be achieved as well as the natural behavior of the structure. With all the simulations carried out through the same numerical model, the differences that could arise in the natural frequencies would be due to the material properties of the piezoelectric layers. Natural frequencies and mode shapes obtained from modal analysis act as the foundation for dynamic analysis.

The six natural frequencies of each piezoelectric material were found and presented in Table 7.4. These frequencies are then employed as the excitation frequencies for the coupled field harmonic analysis.

Table 7.4 : First six natural frequencies for different piezoelectric materials.

Mode	PZT-4 (Hz)	PZT-5A (Hz)	PZT-5H (Hz)	PZT-8 (Hz)	PZT-26 (Hz)
Mode 1	53.065	50.575	50.858	53.494	54.193
Mode 2	289.8	280.72	282.56	290.84	292.77
Mode 3	294.34	282.32	284.14	296.42	299.8
Mode 4	445.64	427.14	431.33	450.4	454.27
Mode 5	707.67	680.64	685.02	711.86	719.05
Mode 6	1145.2	1107	1114.5	1150.3	1160.3

As seen in Table 7.4, the natural frequencies increase successively from Mode 1 to Mode 6 for all tested piezoelectric materials, which is consistent with the theoretical behavior of a cantilever wing. Small changes in the natural frequencies of the modes are evident because of the difference in stiffness and mass properties of the tested piezoelectric materials. The material having the highest value of first natural frequency has a higher dynamic stiffness, while the material having the lowest value of first natural frequency is more flexible in free vibration. Modal analysis gives the natural frequencies and their respective mode shapes. The mode shapes are depicted in Figure 7.4 and provide a visual representation of the vibration pattern of the smart wing. Since the geometry and boundary conditions are the same for all types of materials, the mode shapes will be the same and only the natural frequencies will differ. The

mode shapes shown above are just the relative deformation shapes of the structure and not the physical displacement values since the contour amplitude is varied for visualization purposes.

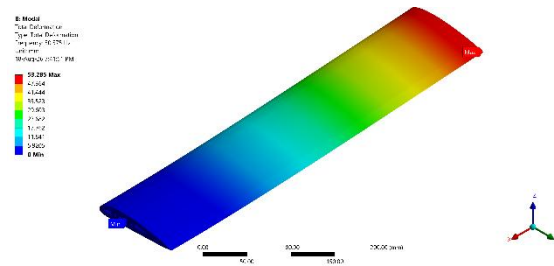


Figure 7.4 : Representative Mode Shapes of the Smart Wing

As seen from Figure 7.4, the lower-order modes predominantly show global bending behavior of the intelligent wing, while the higher-order modes consist of complicated deformations involving both bending and twisting. As the geometry, boundary conditions, and meshing are all the same for the analysis, the mode shapes are also similar for different materials; the main distinction lies in the natural frequencies of each particular case. The given mode shapes show the relative vibration mode shapes of the structure, with the deformation magnitude exaggerated for clarity. From the practical point of view, high natural frequencies are preferable since they will decrease the probability of resonance in real life and increase the structural dynamics stability. Nevertheless, the final choice of the best piezoelectric material cannot be made by only considering modal properties since the other important features such as electromechanical performance and energy harvest potential should be taken into account. The analysis is given in the next chapter.

7.4 Coupled-Field Harmonic Analysis Results

The coupled field harmonic analysis was done in order to investigate the dynamics electromechanical behavior of the proposed smart aircraft wing subjected to harmonic excitation. The analysis will give both the structural response as well as the electrical potential of the piezoelectric material embedded in the structure. The coupled field harmonic analysis in ANSYS analyzes the structural and electric field together for the piezoelectric material, hence making it possible to analyze the mechanical and electrical behavior of the structure.

The excitation frequency range is selected based on the modal natural frequency obtained from the modal analysis. For each of the piezoelectric material, the frequency range is 0 Hz up to 1107 Hz including the six natural frequencies as well as five intermediate frequencies to get an accurate frequency response.

The geometry, mesh, bonded contacts, boundary conditions, material properties of aluminum, electrical loading, and solver were kept constant in order to achieve consistency in comparison. The only varying parameter between the different models was the piezoelectric material that was embedded into the structure. The results of the models were obtained in the form of total deformations, equivalent (von Mises) stress, equivalent elastic strain, electrical potential of all bodies, and electrical potential of piezoelectric bodies, respectively.

7.4.1 Total Deformation

The total deformation resulting from the coupled-field harmonic analysis is the dynamic deformation of the smart wing depending on the excitation frequency. Unlike in the case of static structural analysis, the deformation is frequency-dependent and is greatly increased around the natural frequencies due to the occurrence of resonance. The coupled-field harmonic analysis in ANSYS considers both the structural response and the electromechanical coupling associated with the piezoelectric materials. The total deformation obtained for each type of piezoelectric material at different frequencies is presented in Table 7.5.

Table 7.5 : Maximum total deformation from coupled-field harmonic analysis.

Piezoelectric Material	Maximum Total Deformation (mm)
PZT-4	2.9225e-002
PZT-5A	1.7716e-002
PZT-5H	1.6595e-002
PZT-8	3.6246e-002
PZT-26	3.7373e-002

As illustrated in Table 7.5, there exist some differences in terms of the maximum deformation that results from the application of harmonics among the analyzed piezoelectric materials. The least deformation 1.6595e-002 mm was realized in PZT-5H, while the greatest deformation 3.7373e-002 mm was achieved using PZT-26. These variations have been accounted for by the electromechanical and elasticity of the piezoelectric materials incorporated. The frequency responses of the smart wing are shown in Figure 7.5, thus providing an opportunity to compare the dynamic deformation of the smart wing within the chosen frequency range. Harmonic response analysis determines the structural behavior at steady state under frequency sweep.

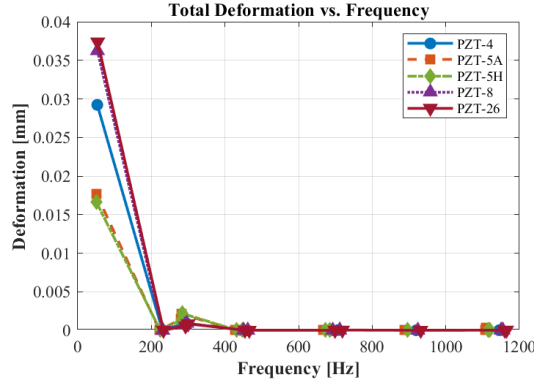


Figure 7.5 : Total deformation frequency response for different piezoelectric materials.

From Figure 7.5, it can be observed that the deformation increases when the excitation frequency gets closer to the natural frequencies of the smart wing, and resonance peaks are created in the frequency response plots. Outside the resonance regions, there is a significant decrease in the deformation, implying that the structure has stable dynamic behavior. In harmonic response analysis, the steady state response of the structure is determined during a frequency sweep, and resonance peaks occur in general at frequencies close to the natural frequencies of the structure.

Figure 7.5 displays the plot for the comparison of frequency response on total deformation. Plots on the frequency response of individual piezoelectric materials have been presented in [Appendix C](#).

Although the total behavior of the piezoelectric materials investigated is almost the same in terms of response plot, the amplitude of the peaks differs since of the variations in their stiffness and electromechanical coupling properties. Piezoelectric materials having smaller peak deformations have higher dynamic stiffness, but peak deformations alone do not give the information about the optimum material to be used. Besides the deformation, the performance of the material is also related with its electrical potential generation and energy harvestability, which will be explained in next parts.

7.4.2 Equivalent (von Mises) Stress

Equivalent (Von Mises) stress determined from the coupled field harmonic analysis gives the dynamic stress generated in the intelligent wing in the chosen frequency range of excitation. Contrary to static structural analysis, the stress changes as a function of excitation frequency and attains its highest value around the resonance frequencies. The coupled field harmonic analysis carried out using ANSYS software analyses both the structure and electrical response of the piezoelectric materials, making it possible to analyze both dynamic stress and generated

voltage in one sweep of the frequency. The maximum equivalent (Von Mises) stress determined for all piezoelectric materials is shown in table Table 7.6 below.

Table 7.6 : Maximum equivalent stress from coupled-field harmonic analysis.

Piezoelectric Material	Maximum Equivalent Stress (MPa)
PZT-4	5.7478
PZT-5A	5.9043
PZT-5H	5.6596
PZT-8	6.4484
PZT-26	5.956

Table 7.6 illustrates slight differences in the peak stress value of the harmonic equivalent stress among the considered piezoelectric materials. While the minimum value of 5.6596 MPa was observed for PZT-5H, the maximum value of 6.4484 MPa was observed for PZT-8. These slight differences were due to the differences in the mechanical and electrical properties of the embedded piezoelectric materials. Frequency response plots are illustrated in Figure 7.6 and will give direct comparisons of the dynamic response of the materials over the excitation frequency range of interest. Harmonic response analysis involves assessing the stress response at the steady state under frequency sweeps.

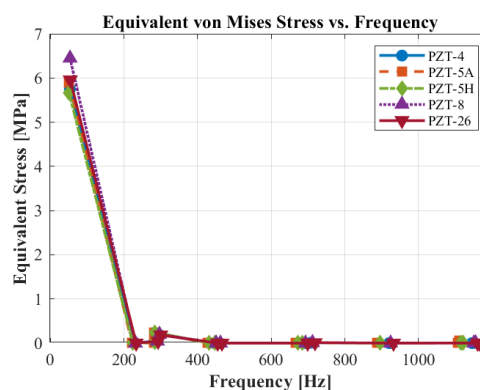


Figure 7.6 : Equivalent stress frequency response for different piezoelectric materials.

According to the graph presented in Figure 7.6, the level of equivalent stress grows with excitation frequency close to the resonance frequencies of the system, resulting in clearly identifiable stress peaks on the frequency responses graphs. With deviation from the resonance frequencies, the stress levels reduce dramatically, suggesting the stable structural response throughout most of the operation range of frequencies. Harmonic response analysis helps to

determine the steady-state stress response in case of the frequency sweep. The peaks of stresses usually occur close to the resonance frequencies. The equivalent stress frequency response plot is depicted in Figure 7.6. Frequency responses for each piezoelectric material individually are included in [Appendix C](#).

The response pattern is almost identical for all the tested piezoelectric materials, but the value of peak stresses varies due to differences in their elastic and electromechanical characteristics. The peak stresses remain at the wing root which is immovable, just like in the static structure analysis of the smart wing. If the maximum equivalent stress for each piezoelectric material does not exceed the yield point of the aluminium alloy (280 MPa), the smart wing is safe to use in the elastic regime.

7.4.3 Equivalent Elastic Strain

Equivalent elastic strain calculated using coupled-field harmonic analysis is indicative of the dynamic elastic behavior of the intelligent wing within the selected frequency range of excitation. As in case with the harmonic stress analysis, the equivalent elastic strain is dependent on the excitation frequency, reaching its peak at resonant frequency values. The evaluation of the equivalent elastic strain is performed by ANSYS using the coupled-field solution method, making the comparison of the elastic behavior of various piezoelectric materials possible. Maximum values of the equivalent elastic strain for all studied piezoelectric materials are listed in Table 7.7.

Table 7.7 : Maximum equivalent elastic strain from coupled-field harmonic analysis.

Piezoelectric Material	Maximum Equivalent Elastic Strain (mm/mm)
PZT-4	9.1112e-005
PZT-5A	6.0971e-005
PZT-5H	5.7432e-005
PZT-8	1.1134e-004
PZT-26	1.1308e-004

The results of comparison, shown in Table 7.7, reveal small differences in the maximal equivalent elastic strain in the investigated piezoelectric materials. The minimal strain of 5.7432e-005 was observed in PZT-5H, whereas the maximal strain of 1.1308e-004 was observed in PZT-26. These differences are due to the elastic and electromechanical behavior

of the incorporated piezoelectric materials. The frequency responses of the materials are shown in Figure 7.7, which makes it possible to compare the dynamic strain behavior in the chosen excitation frequency interval. The equivalent elastic strain can be calculated using the coupled-field harmonic solution, thus comparing the elastic behavior of various material systems in frequency interval.

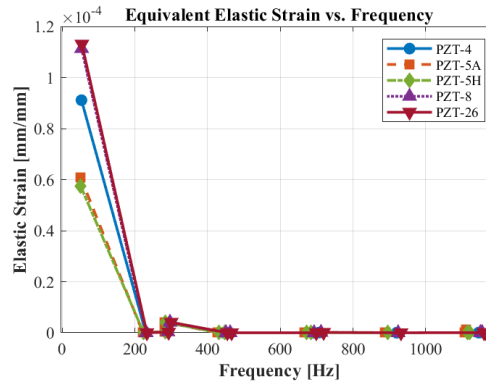


Figure 7.7 : Equivalent strain frequency response for different piezoelectric materials.

As depicted in Figure 7.7, the equivalent elastic strain is increasing with respect to the increase in the excitation frequency close to the resonance frequencies, which results in the peaks in the strain values around the resonance frequencies and reduction in the strain value away from these frequencies. Such behavior is in accordance with the dynamic behavior of a cantilever system subjected to harmonic excitations. Figure 7.7 shows the frequency response plot for equivalent elastic strain. The individual frequency response plots for the piezoelectric materials are given in [Appendix C](#).

While there is no much difference between the general trend of the response of all studied piezoelectric materials, there exists some difference in the peak value of the strain due to the differences in their mechanical and electromechanical properties. Higher dynamic stiffness or lower peak strain value means better resistance to elastic deformation. But in addition to that, the optimal material among those should be selected based on the electrical potential generated and the performance in energy harvesting, since this is the objective of the proposed wing. The equivalent elastic strain in ANSYS is calculated with the help of the elastic strain field by a von Mises formulation.

7.4.4 Electrical Potential

The electrical potential that is generated during coupled-field harmonic analysis denotes the voltage that is generated by piezoelectric materials embedded in the structure via direct piezoelectric effect at harmonic excitation. The voltage thus generated is dependent on the

excitation frequency and attains its maximum value around the resonant frequencies where the structural vibration attains maximum value. Coupled field harmonic analysis in ANSYS is an approach whereby both the structural and electrical fields are solved simultaneously and hence enable frequency dependence of voltage generated by piezoelectric materials to be analyzed. The maximum value of the electrical potential attained by each piezoelectric material, taking into consideration the whole model, has been shown in Table 7.8.

Table 7.8 : Maximum electrical potential from coupled-field harmonic analysis.

Piezoelectric Material	Maximum Electrical Potential (mV)
PZT-4	6111.1
PZT-5A	5432.1
PZT-5H	4185
PZT-8	6275.1
PZT-26	6117.6

As seen from the comparison provided in Table 7.8, the electrical potential generated differs between the investigated piezoelectric materials due to the different values of their piezoelectric constants and electromechanical coupling parameters. The maximum electrical potential was found to be 6275.1 mV for PZT-8 and the minimum electrical potential to be 4185 mV for PZT-5H. The frequency-response graphs of these materials are shown in Figure 7.8. Thus, the direct comparison of the voltage generated at different excitation frequencies is possible. Harmonic coupled-field analysis implemented in ANSYS allows obtaining frequency-dependent voltage results for piezoelectric materials.

Figure 7.8 shows the comparative graph of electrical potential with respect to all the materials. Frequency responses graphs for each individual piezoelectric material have been given in [Appendix C](#).

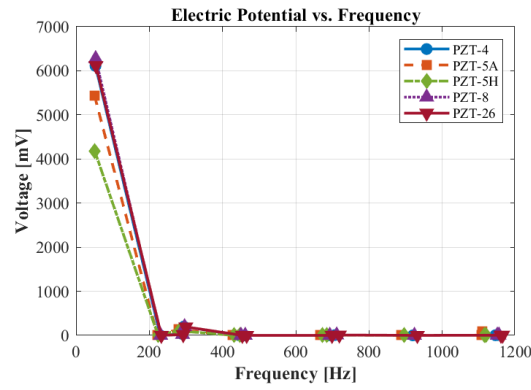


Figure 7.8 : Electrical potential frequency response for different piezoelectric materials.

As illustrated in Figure 7.8, the electrical potential rises substantially as the excitation frequency gets closer to the resonance frequencies because the potential reaches its peak value at resonance when the vibration amplitude is maximum. Otherwise, the produced voltage will drop since the response of the structure is less noticeable. Voltage response is one of the indicators of performance of energy harvesting due to inherent nature of coupled-field harmonic analysis which is based on the electromechanical coupling of the piezoelectric material.

Although all materials tested have the same trend in frequency response, there are certain differences among the values of peak electrical potential owing to the different values of piezoelectric coefficients and electromechanical coupling of the materials. Materials which produce larger voltage under the same excitation conditions indicate that they are more appropriate for vibration energy harvesting and therefore can be considered as optimum materials for the design of the smart wing.

7.5 Comparative Evaluation of Piezoelectric Materials

The main goal of this study was to find out the best material that can be used as a piezoelectric in the smart sandwich wing design of the aircraft. This study compares the structural, dynamic and electromechanical performance of the finite element analysis. In order to make the comparisons fair, all analyses were done with the same wing geometry, same aluminum material, same finite element meshing, same contact, loading and electrical boundary condition and solver settings. The only variable in the study is the piezoelectric material.

In order to make an effective comparison, the main numerical outputs from the static structural analysis, modal analysis, and coupled field harmonic analysis have been presented in Table 7.9. These include the maximum static displacement, equivalent stress, equivalent elastic strain, first natural frequency, maximum harmonic displacement, maximum harmonic stress,

maximum harmonic strain, and maximum electrical potential generated by the piezoelectric materials. All key performance parameters being shown in one table makes a proper engineering evaluation possible in contrast to making any engineering evaluation based on only one output. According to the engineering report guidelines, it is advised that more than one performance parameter be considered prior to making any decision on the basis of only one output.

Table 7.9 : Overall Comparison of the Investigated Piezoelectric Materials

Performance Parameter	PZT-4	PZT-5A	PZT-5H	PZT-8	PZT-26
Static Total Deformation (mm)	0.1459	0.1583	0.1588	0.1427	0.1387
Static Equivalent Stress (MPa)	1.8991	2.0372	2.0324	1.849	1.8923
Static Equivalent Elastic Strain (mm/mm)	2.9862e-005	3.2522e-005	3.3532e-005	2.8339e-005	2.7441e-005
First Natural Frequency (Hz)	53.065	50.575	50.858	53.494	54.193
Maximum Harmonic Deformation (mm)	2.9225e-002	1.7716e-002	1.6595e-002	3.6246e-002	3.7373e-002
Maximum Harmonic Stress (MPa)	5.7478	5.9043	5.6596	6.4484	5.956
Maximum Harmonic Elastic Strain (mm/mm)	9.1112e-005	6.0971e-005	5.7432e-005	1.1134e-004	1.1308e-004
Maximum Harmonic Electrical Potential (mV)	6111.1	5432.1	4185	6275.1	6117.6

As can be seen from the comparison shown in Table 7.9, each piezoelectric material possesses its unique ratio of structural properties and capabilities for energy harvesting. Materials that

yield smaller deformations, stresses, and strains possess better rigidity and stability, whereas materials that yield larger electrical potentials are better in terms of energy harvesting capability. Therefore, choosing an optimal material involves taking into account all the response properties and not just one property.

Static structural analysis showed that **PZT-26** possessed the smallest deformation, whereas **PZT-8** showed the minimal equivalent stress and equivalent elastic strain. Thus, these materials should be characterized by better rigidity and reduced loading due to aerodynamic pressure. Similarly, modal analysis showed that **PZT-26** had the largest first natural frequency of **54.193 Hz**.

Coupled field harmonic analysis also confirmed that the frequencies obtained during the modal analysis coincide with the maximum structural response and the maximum electrical potential generated. Among the investigated materials, **PZT-8** has the highest electrical potential of **6275.1 mV** and is therefore the best material in the conversion of mechanical to electrical energy. The **PZT-5H** has the least electrical potential and hence has relatively poor energy harvesting capabilities under the same conditions.

While some materials have the maximum electrical potential, they may not have the minimum deformations or stresses. Thus, the choice of material should depend on the applications of the smart wing. For instance, when structural stiffness is the goal of the design, then it would be ideal to choose the material that has the least deformation and stress. If vibration energy harvesting is the goal, the best material is that which has the maximum electrical potential. The choice of the optimum material in the current research is thus made on a basis of a combination of these parameters.

Given the numerical outcomes generated through all analyses carried out, **PZT-26** performed the best in terms of overall behavior since it offered the right balance between structural performance and electricity generation. Therefore, **PZT-26** would be the best material to use for the intelligent sandwich aircraft wing.

Chapter 8 Conclusions And Future Scope

8.1 Conclusions

This research was successful in analyzing the behavior of the smart sandwich aircraft wing incorporated with piezoelectric materials in terms of structure and electromechanical aspects using finite element analysis in ANSYS Mechanical 2025 R2. The three-layer sandwich composite was used in this research to understand the effect of piezoelectric materials on the structural performance, dynamic behavior, and vibration energy harvest.

A 3D CAD model of the wing was created in CATIA V5 and was imported in ANSYS for numerical modeling purposes. In order to keep all cases comparable, all other factors such as geometry, meshing, bonded contact, loads, constraints, and material properties of aluminum were maintained constant, while only the embedded piezoelectric material was changed.

Static structural analysis results proved that all studied models met the structural requirements under aerodynamic loads. Deformation, equivalent stress, and equivalent elastic strain values of all structures were within the acceptable values, and the maximum value of stress was less than yield stress of aluminum alloy, which shows that the proposed structure works in an elastic zone.

The successful identification of the six natural modes of frequency and the associated mode shapes of the smart wing by the modal analysis was observed. The results indicated the expected vibration response of a cantilever wing in terms of bending and torsion, but with variations in natural frequencies because of variation in the properties of stiffness and mass of the piezoelectric materials.

The coupled field harmonic analysis was effective in demonstrating the conversion of structural vibration into electric potential using the direct piezoelectric effect. The structural responses and the generated voltage were at their maximum near the resonant frequencies, indicating the high correlation between structural vibration and energy generation. Harmonic coupled field analysis in ANSYS is a simultaneous solution of the structural and electric fields.

Overall, the thorough comparison of all the results revealed that the best piezoelectric material should be such a one that would be the compromise between stiffness, dynamics, and the capability to harvest energy, rather than have the best value of a certain type of the response. On the basis of numerical results obtained from this analysis, **PZT-26** was found to be the most efficient and, therefore, recommended for the development of smart sandwich aircraft wing.

Thus, from the point of view of the results obtained from the conducted analysis, the use of the piezoelectric materials in a sandwich aircraft wing is a promising concept for the development of efficient smart aerospace structures that will be able to retain their stability along with energy harvesting from the vibrations.

8.2 Limitations of the Present Study

However, despite the successful demonstration of the behavior of the structure of the proposed smart sandwich aircraft wing, the results obtained need to be discussed with some reservations due to the following aspects.

The present research is completely done via numerical simulation using ANSYS Mechanical 2025 R2 software and there are no experimental verification results available. The latter is one of the ways to provide further support to the obtained results. Moreover, the numerical analysis made in the present work is based on the assumption that all materials are behaving elastically, which means that no practical aspects like the impact of material aging, manufacturing defects, adhesive effects and others were included in the analysis. The analysis carried out in ANSYS is based on the assumptions of linear behavior of materials.

The aerodynamic load case was modeled using uniform pressure distribution with fixed support at the wing root to simulate the cantilever action. While this ensures consistency in comparison, aircraft wings experience continuously changing aerodynamic loads based on flight speed, altitude, maneuvers, and angle of attack. Also, the coupled-field harmonic analysis was done only up to the frequency range from the first six natural frequencies, but no higher vibration mode or transient effect was considered. In ANSYS, harmonic analysis is used to study the steady-state vibration of linear structures and cannot consider transient vibration behavior.

The study was limited in scope as it compared only five different commercially available piezoelectric materials, keeping the wing geometry, mesh, boundary conditions, and structural configurations the same. Other design parameters, such as the thickness of piezoelectric layers, patch size, patch locations, stacking order, electrical load resistance, and power-conditioning circuits, were deliberately kept constant.

Despite all these restrictions, the chosen numerical approach offers a good basis for the estimation of smart sandwich wings reinforced with piezoelectric materials. The results obtained can be considered to be a good starting point for the further experiments and research in terms of real-life wing loading and nonlinearity.

8.3 Future Scope

This current research indicates the viability of embedding piezoelectric materials into a smart aircraft wing in order to harvest energy by means of vibrations. Even though the research has been performed only numerically, there are a number of ways to develop the current research further.

First of all, the fabrication of the smart wing proposed and its experimental testing must be done in order to validate the results of the numerical simulation. Moreover, the flight conditions can be analyzed with the use of the fluid structure interaction (FSI) and aeroelastic approach.

Further research would involve optimizing the thickness of the piezoelectric layer, its size and location, as well as the order of its layers, in order to enhance the performance of the structure and increase energy harvesting efficiency. Energy storage, power conditioning circuits, and low-power sensors may assist in developing self-powered structural health monitoring systems as well as wireless sensor networks.

This methodology could also be used to explore different smart materials, hybrid piezoelectric composites, and multifunctional sandwich structures. Moreover, by incorporating active vibration control, adaptive structures, and morphing wing technologies with the embedded piezoelectric materials, one may enhance the performance of an airplane while simultaneously creating electrical energy.

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APPENDICES

Appendix A: Clark-Y Airfoil Coordinates

Sl. No	X	Y
1	100	0
2	99	0.2969
3	98	0.53335
4	97	0.76868
5	96	1.00232
6	94	1.46239
7	92	1.91156
8	90	2.35025
9	88	2.77891
10	86	3.1974
11	84	3.60536
12	82	4.00245
13	80	4.38836
14	78	4.76281
15	76	5.12565
16	74	5.47675
17	72	5.81599
18	70	6.14329
19	68	6.45843
20	66	6.76046
21	64	7.04822
22	62	7.32055
23	60	7.57633
24	58	7.81451
25	56	8.0348
26	54	8.23712
27	52	8.42145
28	50	8.58772
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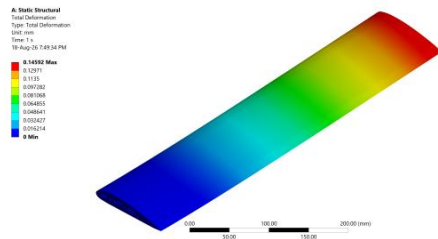
Sl. No	X	Y
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67	1.2	-1.69733
68	2	-2.02723
69	3	-2.26056
70	4	-2.45211
71	5	-2.60452
72	6	-2.71277
73	8	-2.84595
74	10	-2.93786
75	12	-2.99633
76	14	-3.02404
77	16	-3.02546
78	18	-3.0049
79	20	-2.96656
80	22	-2.91445
81	24	-2.85181
82	26	-2.78164
83	28	-2.70696
84	30	-2.63079
85	32	-2.55565
86	34	-2.48176
87	36	-2.4087
88	38	-2.33606
89	40	-2.26341
90	42	-2.19042

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31	44	8.97175
32	42	9.05657
33	40	9.11712
34	38	9.15212
35	36	9.16266
36	34	9.15079
37	32	9.11857
38	30	9.06804
39	28	9.00016
40	26	8.9084
41	24	8.78308
42	22	8.61433
43	20	8.39202
44	18	8.10687
45	16	7.75707
46	14	7.3436
47	12	6.86204
48	10	6.29981
49	8	5.64308
50	6	4.87571
51	5	4.42753
52	4	3.91283
53	3	3.30215
54	2	2.53735
55	1.2	1.78581
56	0.8	1.3735
57	0.4	0.89238
58	0.2	0.58025
59	0.1	0.37271
60	0.05	0.2339
61	0	0

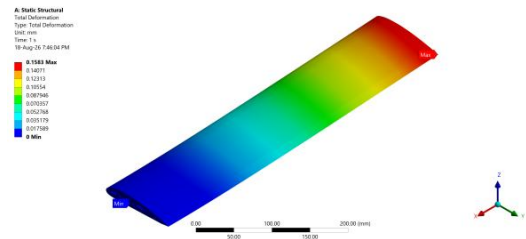
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98	58	-1.60232
99	60	-1.52893
100	62	-1.45551
101	64	-1.38207
102	66	-1.30862
103	68	-1.23515
104	70	-1.16169
105	72	-1.08823
106	74	-1.01478
107	76	-0.94133
108	78	-0.86788
109	80	-0.79443
110	82	-0.72098
111	84	-0.64753
112	86	-0.57408
113	88	-0.50063
114	90	-0.42718
115	92	-0.35373
116	94	-0.28028
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Appendix B: Static Structural Contour Plots

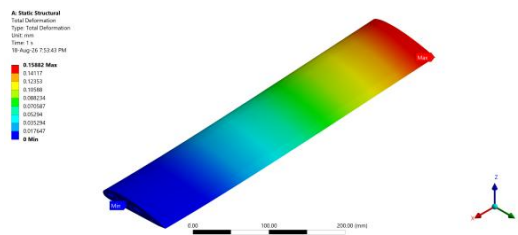
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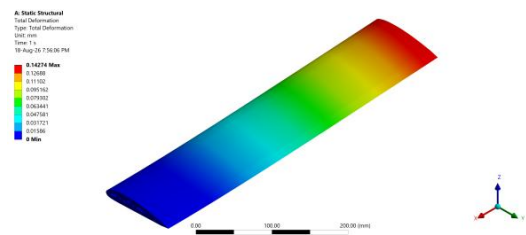
PZT - 4



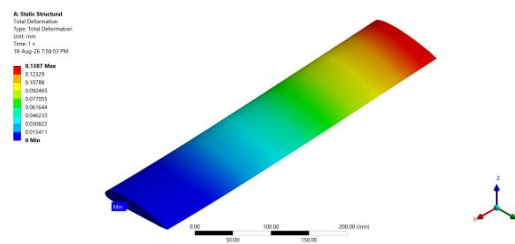
PZT - 5A



PZT - 5H

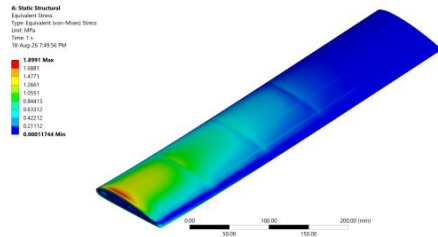


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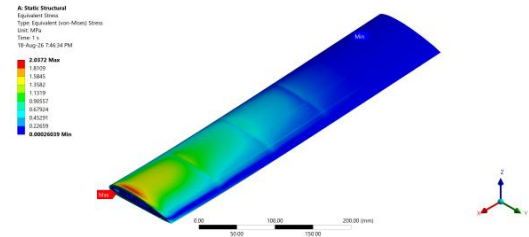


PZT - 26

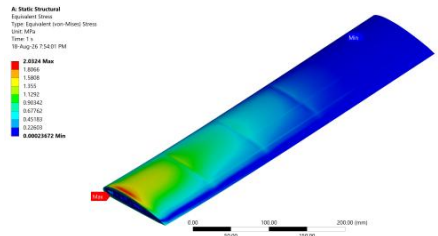
B.2 Equivalent Stress



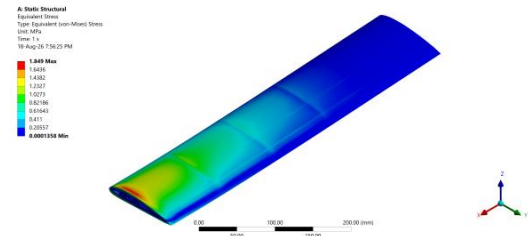
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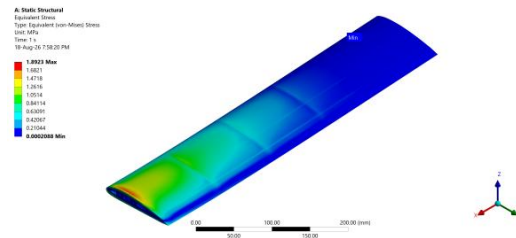
PZT - 5A



PZT - 5H

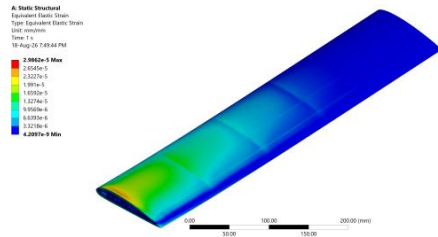


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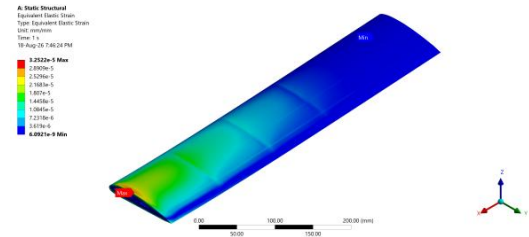


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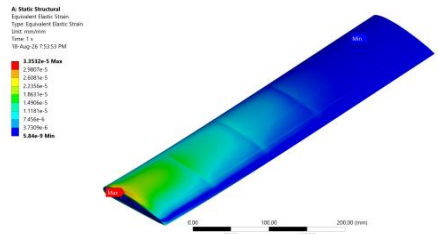
B.3 Equivalent Strain



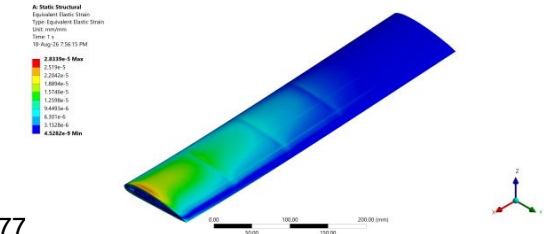
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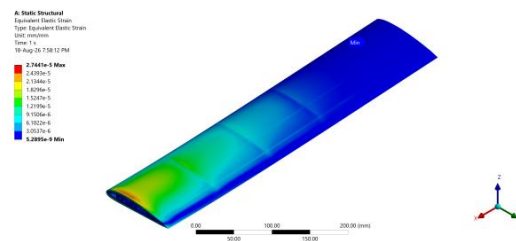
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PZT - 5H



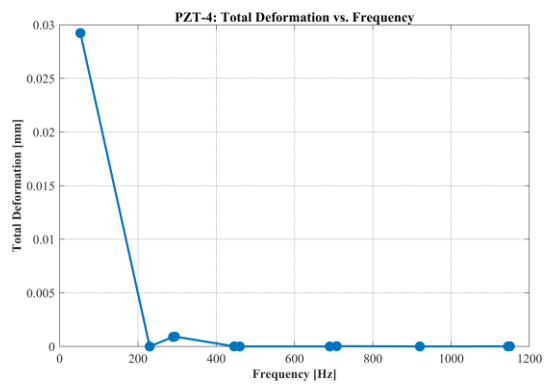
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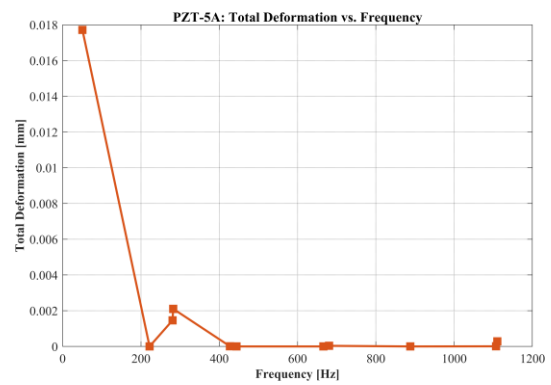
PZT - 26

Appendix C: Coupled Field - Harmonic Response Frequency Plots

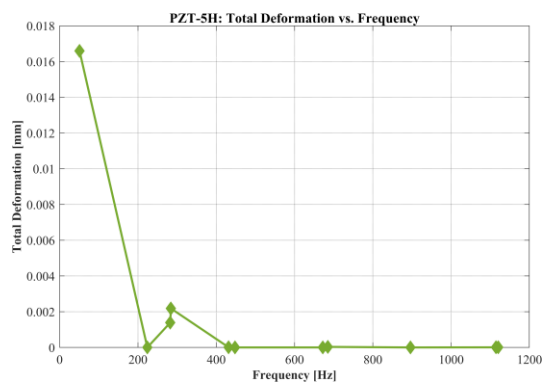
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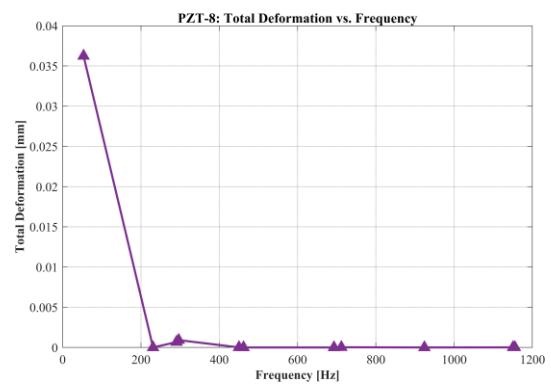
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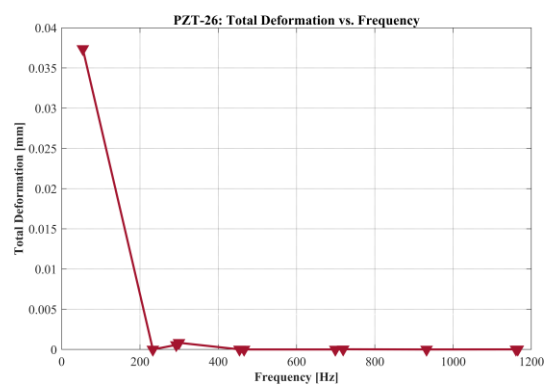
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PZT - 5H

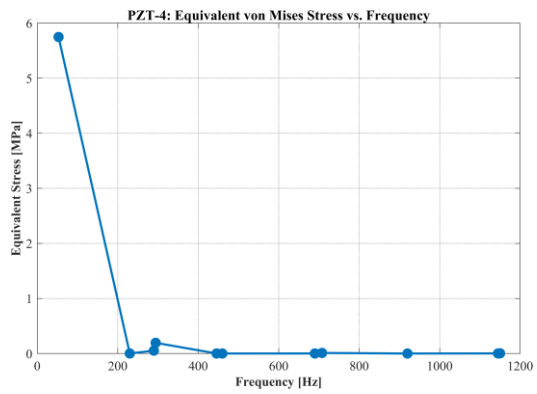


PZT - 8

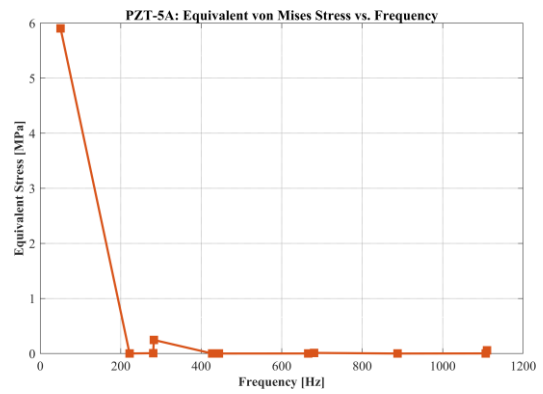


PZT - 26

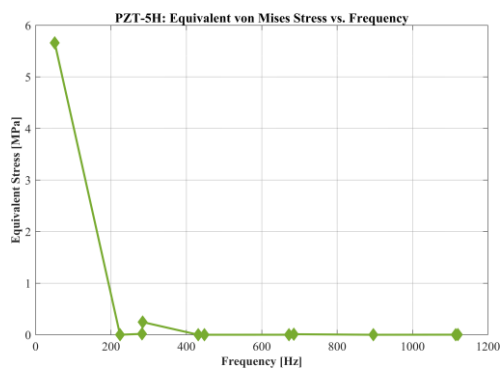
C.2 Stress Plots



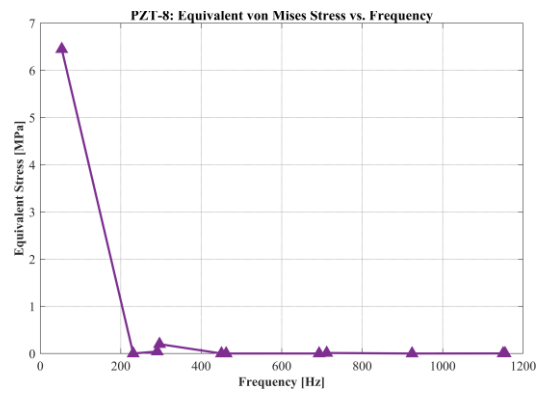
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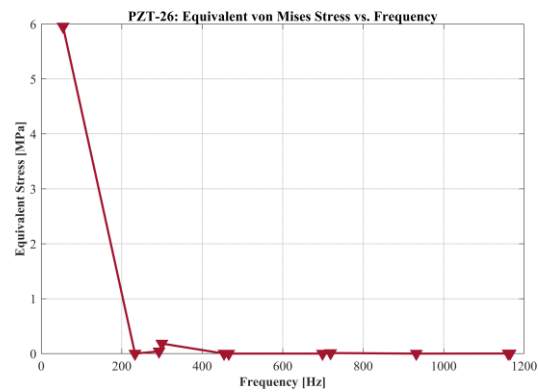
PZT - 5A



PZT - 5H

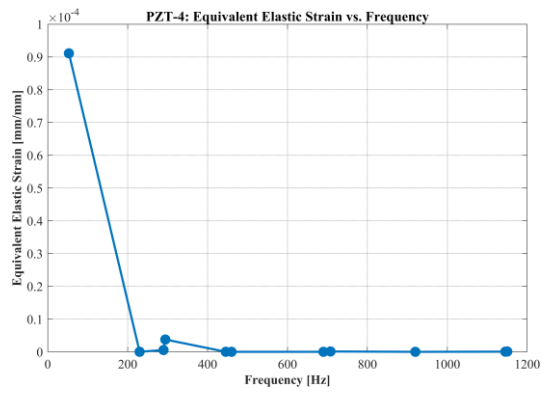


PZT - 8

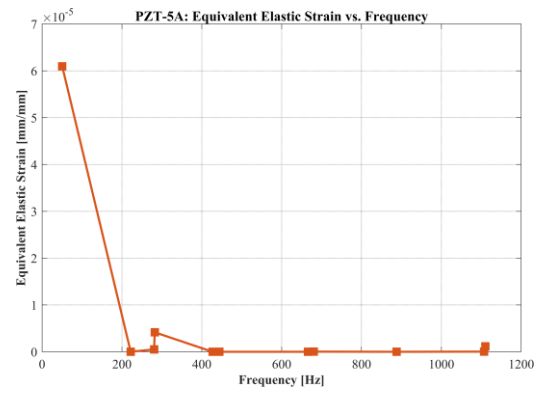


PZT - 26

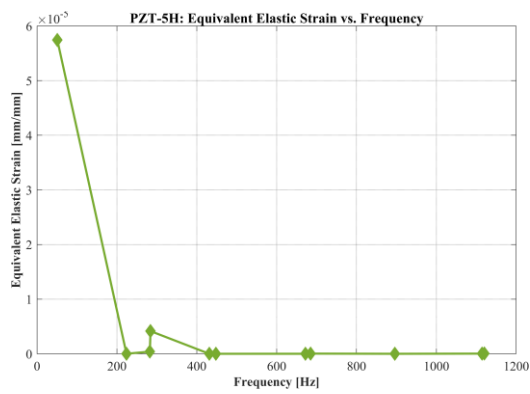
C.3 Strain Plots



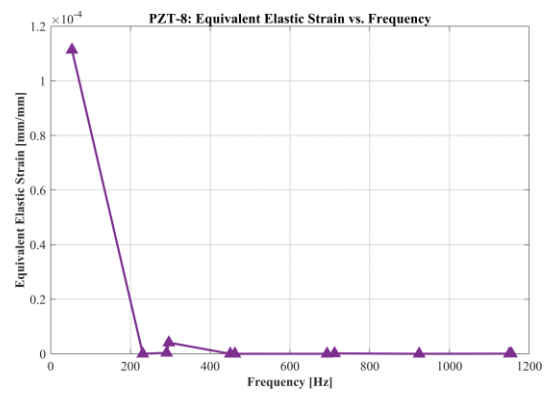
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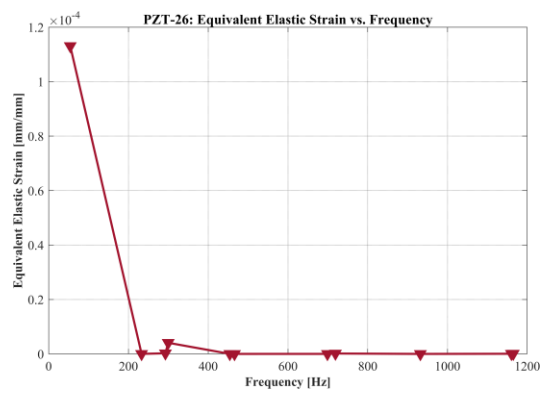
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PZT - 5H

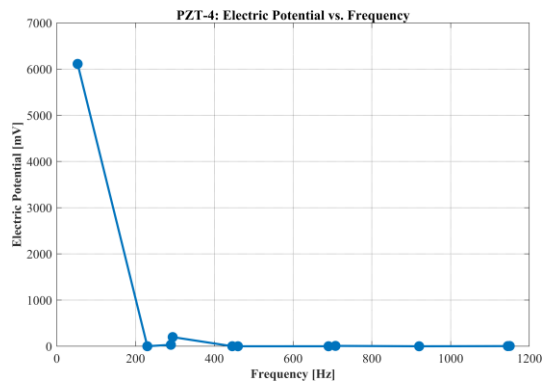


PZT - 8

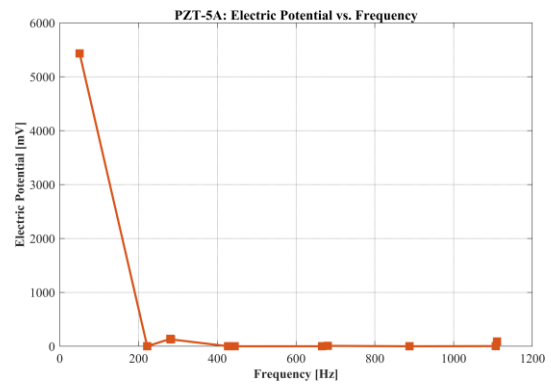


PZT - 26

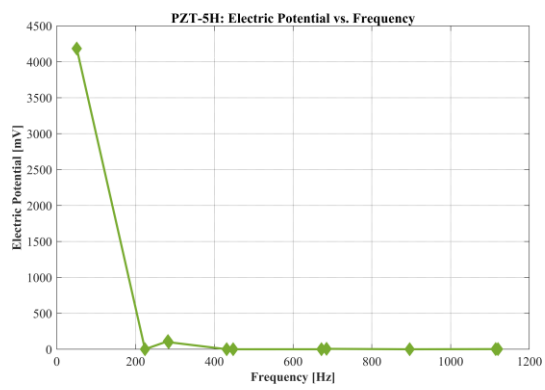
C.4 Electric Potential Plots



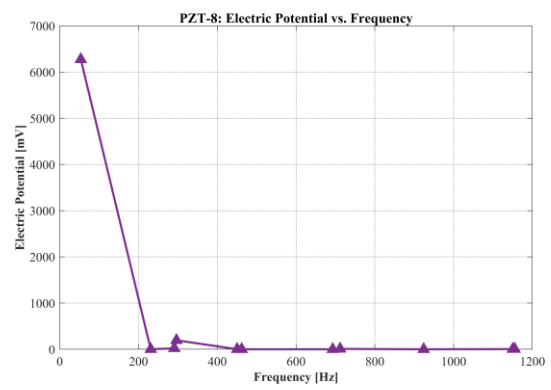
PZT - 4



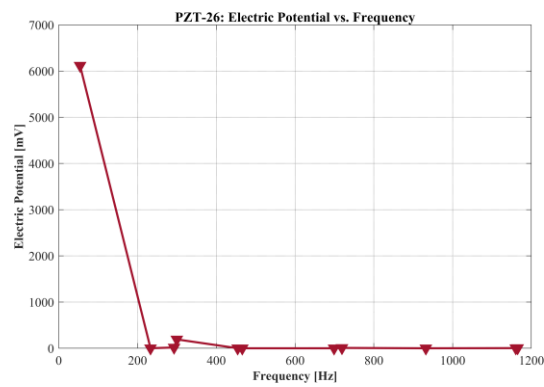
PZT - 5A



PZT - 5H



PZT - 8



PZT - 26